Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

Emma Bach

2026-07-19

Contents

Part I

Foundations

Chapter 1

An Introduction to Mathematical Reasoning

Chapter 2

Sets

Almost all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "b is an element of M", or equivalently "b contains M", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.

2. The number of times an element appears within the same set doesn't matter. $\{1\}, \{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

2.7 Replacement

2.8 Regularity and Well-Foundedness

2.9 Choice

2.10 Cardinality

Part II

Algebraic Structures

Chapter 1

$(\mathbb{N}, +)$ is a Monoid

Chapter 2

$(\mathbb{Z}, +, -)$ is a Group

2.1 Group Homomorphisms

Theorem 2.1. *Let $\varphi : (G, \circ_G) \to (H, \circ_H)$ be a map that respects group operations, i.e:*

$$\varphi(g_1 \circ_G h_2) = \varphi(g_1) \circ_H \varphi(g_2)$$

Then h also respects the neutral element and inverses, i.e:

$$\begin{aligned} \varphi(e_G) &= e_H \\ \varphi(g^{-1}) &= \varphi(g)^{-1} \end{aligned}$$

Proof. 1. For any $h \in H$, we have:

$$\begin{aligned} e_H &= e_H \circ_H e_H \\ &= \left(\varphi(e_G)^{-1} \circ_H \varphi(e_G)\right) \circ_H e_H \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G \circ_G e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \circ_H \varphi(e_G) \\ &= e_H \circ_H \varphi(e_G) \\ &= \varphi(e_G) \end{aligned}$$

So φ respects the neutral element.

2. Since φ respects the neutral element, we get:

$$\begin{aligned} \varphi(g) \circ_H \varphi(g^{-1}) &= \varphi(g \circ_G g^{-1}) \\ &= \varphi(e_G) \\ &= e_H, \end{aligned}$$

i.e. $\varphi(g^{-1}) = \varphi(g)^{-1}$.

□

2.2 Group Actions

Definition 2.2

Let $(G, \circ, e)$ be a group. Let X be a set. Then a map $. : G \times X \to X$ is called a ***group action*** of G on X iff:

1. It respects the neutral element:

$$\forall x \in X :$$
$$e.x = x$$

2. It associates with the group operation:

$$\forall g, h \in G, x \in X :$$
$$(g \circ h).x = g.(h.x)$$

Exercise 2.3. *Show that a map $(g, x) \to g.x$ is a group operation if and only if the map $g \to (x \to g.x)$ is a group homomorphism from G into the symmetric group over X.*

Definition 2.4. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$Gx := \{g.x \mid g \in G\}$$

of all points that x can be taken to by an element of G the ***orbit*** of x under G.

Exercise 2.5. *Let $X = \mathbb{R}^2$ be the plane. Let G be the group of rotations around the origin. Sketch the orbits of different points $x \in X$.*

Definition 2.6. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the set

$$X/G := \{Gx \mid x \in X\}$$

of all orbits under G the ***orbit space*** of G in X.

Exercise 2.7. *Show that the relation*

$$x \sim y \Leftrightarrow \exists g \in G : g.x = y$$

is an equivalence relation. Show that for any $x \in X$, we the equivalence class of x is exactly the orbit of x.

Definition 2.8. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the action . ***transitive*** if any point $x \in X$ can be taken to any point $y \in X$ by an element of G:

$$\forall x, y \in X :$$
$$\exists g \in G :$$
$$g.x = y$$

Exercise 2.9. *Show that a group action is transitive if and only if $Gx = X$ for all $x \in X$.*

Definition 2.10. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$G_x := \{g \in G \mid g.x = x\}$$

of all elements of G leaving x unchanged the ***isotropy group*** of x.

Exercise 2.11. *Show that the isotropy group G_x is always a subgroup of G.*

Exercise 2.12. *Show that if the action $(g, x) \mapsto g.x$ is transitive, the map*

$$\begin{aligned} \varphi_x : G/G_x &\to X \\ [g] &\mapsto g.x, \end{aligned}$$

where G/G_x is the quotient group of G by G_x, is well-defined and bijective.

We can also define the isotropy groups of points:

Definition 2.13. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $A \subseteq X$. Then we define

$$g.A := \{g.a \mid a \in A\}$$

and call the set

$$G_A := \{g \in G \mid g.A = A\}$$

of all elements of G mapping elements of A to elements of A the ***isotropy group*** of A.

Chapter 3

$(\mathbb{Z}, +, -, \cdot)$ is a Ring

Chapter 4

$(\mathbb{Q}, +, -, \cdot, ^{-1})$ is a Field

Part III

Foundations of Analysis

Chapter 1

The Real Numbers $\mathbb{R}$

1.1 Ordered Fields

1.2 Dedekind Cuts

1.3 Uniqueness of $\mathbb{R}$

Chapter 2

The Concept of Distance

2.1 Metric Spaces

2.2 Convergence

2.3 Continuity

Chapter 3

Infinite Series

Chapter 4

Euclidean Geometry

4.1 The Complex Numbers $\mathbb{C}$

4.2 Circles, Angles and Trigonometry

Chapter 5

Function Spaces

Chapter 6

Differential Calculus in One Variable

Chapter 7

The Riemann-Stieltjes Integral

Part IV

Linear Algebra

Chapter 1

Analytic Geometry

1.1 Arithmetic in n dimensions

We now focus our attention on n-dimensional euclidean space, which is modelled by the cartesian product $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\} = \left\{ \vec{x} : \{1, \ldots, n\} \to \mathbb{R} \right\}$$

Elements of $\mathbb{R}^n$ are known as ***vectors***[1]. In the context of linear algebra, they are usually written vertically, as ***column vectors***.

[1] We will see in the next chapter that $\mathbb{R}^n$ is an example of a ***vector space***, including the case $n = 1$ (and $n = 0$, if you want to get a bit pathological). Therefore, formally speaking and only in certain contexts, regular old real (or even rational) numbers can be vectors too. However, when encountering the term "vector" in the wild, it is generally implied that $n \geq 2$.

Vectors simultaneously represent both ***positions*** and ***directions*** in space - in the same way that the number 5 is both the point that lies 5 units away from 0 on the real number line *and* the distance between the numbers 3 and 8 on that same number line, the vector $\binom{1}{2}$ represents both

- The point on the cartesian plane that can be reached by going one unit in the direction of one coordinate axis and then two units in the direction of the other axis, *and*
- the distance between two points with coordinates $\binom{2}{3}$ and $\binom{3}{5}$, or any other pair of points that are one unit apart in one coordinate direction and two in the other.

As is expected for algebraic structures defined on cartesian products, addition is defined **componentwise**:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Given a real number c, we can scale elements of $\mathbb{R}^n$ by that number:

$$c \cdot \vec{x} = c \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{pmatrix}$$

The factor c is known as a ***scalar***, and the operator $\cdot : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ as ***scalar multiplication***. For the sake of readability, we differentiate scalars from vectors by writing a little arrow on top of a vector.

Arguably, it is these two operations - vector addition and scalar multiplication - that define linear algebra - next chapter, we will abstractly define ***vector spaces*** to be those structures where addition and scalar multiplication are well-behaved and the scalars come from a field, and ***modules*** to be their cousins whose scalars come from a ring.

In general, there is no well-behaved[2] vector multiplication that works for any n. Later on, we will prove this observation in the form of ***Frobenius' Theorem***, which roughly states:

Theorem 1.1: Frobenius' Theorem

The only dimensions in which an associative multiplication exists on $\mathbb{R}^n$ are $\mathbb{R}^1$, i.e. $\mathbb{R}$ itself, $\mathbb{R}^2$, where multiplication is given by treating two-dimensional vectors as complex numbers, and $\mathbb{R}^4$, where multiplication is given treating four-dimensional vectors as ***quaternions***.

We will take a closer look at the complex numbers $\mathbb{C} \cong \mathbb{R}^2$ and the quaternions $\mathbb{H} \cong \mathbb{R}^4$ in the following sections. In particular, we will see that quaternion multiplication isn't commutative, so if we require multiplication to be commutative, our options shrink even further, leaving just $\mathbb{R}$ and $\mathbb{C} \cong \mathbb{R}^2$.

[2] In this case, our criterion for multiplication being well-behaved is that multiplicative inverses should exist, i.e. that division should be possible.

1.2 The Inner Product: Angles and Lengths in $\mathbb{R}^n$

Another operation that is of central importance to euclidean geometry is the ***inner product***, also known as the ***dot product*** or ***scalar product***.

Definition 1.2: Euclidean Inner Product

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, their ***Euclidean inner product***, written $\langle \vec{x}, \vec{y} \rangle$ or simply $\vec{x} \cdot \vec{y}$, is the sum of the products of the components of $\vec{x}$ with the components of $\vec{y}$:

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

Theorem 1.3: Properties of Inner Products

The Euclidean inner product has the following properties:

1. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, we have $\langle \vec{x}, \vec{x} \rangle = 0$ if and only if $\vec{x} = \vec{0}$.
2. ***(Linearity in the first argument):*** For every $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, we have:

$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$$

3. ***(Symmetry):*** For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have:

$$\langle x, y \rangle = \langle y, x \rangle$$

Exercise 1.4. *Show that the Euclidean inner product fulfills the following version of the binomial formula:*

$$\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$
$$\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle - 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$

The importance of inner products comes from the fact they let us define ***lengths*** and ***angles***.

Definition 1.5: Euclidean Norm

The length of a vector $\vec{x} \in \mathbb{R}^n$, also known as its ***Euclidean norm***, written as $\|\vec{x}\|_2$, $\|\vec{x}\|$, or just $|x|$, is given by:

$$\|\vec{x}\|_2 = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \sqrt{x_1^2 + \ldots + x_n^2}$$

Note that for $n = 1$, we simply get $\|x\|_2 = \sqrt{x^2} = |x|$, so any result concerning the Euclidean norm on $\mathbb{R}^n$ also holds for the absolute value function.

Theorem 1.6: Some properties of Norms

The Euclidean norm has the following properties:

1. ***(Non-negativity):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 \geq 0$.

2. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 = 0$ if and only if $\vec{x} = \vec{0}$.

3. ***(Absolute homogeneity):*** For every $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, we have

$$\|c \cdot \vec{x}\|_2 = |c| \cdot \|\vec{x}\|_2$$

The little subscript 2 in $\|\vec{x}\|_2$ differentiates the Euclidean norm from the many other norms we will meet over time - in general, for any $p \in \mathbb{N}_1$, $\|\vec{x}\|_p$ denotes the function

$$\|\vec{x}\|_p = \sqrt[p]{|x_1|^p + \ldots + |x_n|^p} \qquad \left(= \sqrt[p]{\sum_{i=1}^{n} |x_i|^p} \right),$$

known as the ***p-norm***. For now, we only really care about the Euclidean norm, but other values of p will start coming in handy once we start applying tools from analysis to vector spaces. The limiting case $p \to \infty$ is particularly useful: We will find that

$$\|\vec{x}\|_\infty = \max |x_i|.$$

Theorem 1.7: Cauchy-Schwartz Inequality

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then we have

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \cdot \|\vec{y}\|_2$$

Proof.

1. Assume $\|\vec{x}\|_2 = 0$. Then we immediately get

$$\langle \vec{x}, \vec{y} \rangle = 0 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2.$$

2. Assume $\|\vec{x}\|_2 = \|\vec{y}\|_2 = 1$. Then we have $\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2 = 2$, and thus:

$$\begin{aligned} 1 + \langle \vec{x}, \vec{y} \rangle &= \frac{1}{2}(\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2) + \langle \vec{x}, \vec{y} \rangle \\ &= \frac{1}{2}(\|\vec{x}\|_2^2 + 2\langle \vec{x}, \vec{y} \rangle + \|\vec{y}\|_2^2) \\ &= \frac{1}{2}\|\vec{x} + \vec{y}\|_2^2 \\ &\geq 0 \end{aligned}$$

Thus we get $\langle \vec{x}, \vec{y} \rangle \geq -1$. Applying the same reasoning to $1 - \langle \vec{x}, \vec{y} \rangle$, we get

$$1 - \langle \vec{x}, \vec{y} \rangle = \frac{1}{2}\|\vec{x} - \vec{y}\|_2^2 \geq 0,$$

i.e. $\langle \vec{x}, \vec{y} \rangle \leq 1$. We thus have

$$|\langle \vec{x}, \vec{y} \rangle| \leq 1 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2,$$

as desired.

3. Now, let $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$ be arbitrary. Then $\frac{\vec{x}}{\|\vec{x}\|_2}$ and $\frac{\vec{y}}{\|\vec{y}\|_2}$ both have norm 1, letting us simply use our previous result:

$$\begin{aligned}
\left|\langle \vec{x}, \vec{y} \rangle\right| &= \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\langle \frac{\vec{x}}{\|\vec{x}\|}, \frac{\vec{y}}{\|\vec{y}\|} \right\rangle \right| \\
&\leq \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\| \frac{\vec{x}}{\|\vec{x}\|} \right\| \cdot \left\| \frac{\vec{y}}{\|\vec{y}\|} \right\| \right| \\
&= \|\vec{x}\| \cdot \|\vec{y}\|
\end{aligned}$$

□

Corollary 1.8: Triangle Inequality

For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have

$$\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$$

We had previously formalized the concept of an 'angle between two lines' via the complex exponential function, and then defined the trigonometric functions $\sin x$ and $\cos x$ as the complex and real parts of e^{ix}, respectively. If we take the cosine function as a given, then scalar products provide a convenient alternate way of defining angles, which works in any dimension:

Definition 1.9: Inner angle between two vectors

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$, we define the ***inner angle*** between them to be:

$$\measuredangle(\vec{x}, \vec{y}) = \arccos\left(\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \right)$$

Note that Cauchy-Schwartz gives us $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \in [-1, 1]$, showing that this expression actually is well-defined.

Exercise 1.10. *Verify that we have:*

1. $\measuredangle(\vec{x}, \vec{y}) = 0$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = c\vec{y}$.
2. $\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2}$ *if and only if* $\langle x, y \rangle = 0$.
3. $\measuredangle(\vec{x}, \vec{y}) = \pi$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = -c\vec{y}$.

Notably, this suggests that our definition is sensible even in the case $n = 1$, since any pair of non-zero vectors fulfills either condition 1 or 2, implying that the angle between two vectors on the real number line is either 0 or π.

These inner angles match our earlier definition of angles in terms of complex numbers, as long as we are content with not being able to meaningfully talk about negative angles or angles greater than π anymore:

Exercise 1.11. *Earlier in this book, we defined angles in terms of complex numbers. Given a complex number* $y = re^{i\varphi}$ *we saw that the corresponding real vector is given by* $\vec{y} = \binom{r\cos\varphi}{r\sin\varphi}$.

Show that our two definitions match as long as $\varphi \in [0, \pi]$, *i.e. that given a vector* $\vec{x} = \binom{x}{0}$ *lying on the x-axis, we have*

$$\measuredangle(\vec{x}, \vec{y}) = \varphi$$

(Hint: Remember $\cos^2\varphi + \sin^2\varphi = 1$*)*

We have to be content with this restricted interval $[0, \pi]$, since the cosine function isn't injective on larger intervals, meaning the arccosine would no longer be well-defined.

More generally, any definition of an 'angle between two arbitrary vectors' will suffer from this - if we only specify two vectors, it's left ambiguous which side we should measure the angle on.

This problem can be solved by finding a way to take a tuple $(\vec{x}_1, \dots, \vec{x}_n)$ of vectors and recognize whether it is 'positively ordered' or 'negatively ordered' - in $\mathbb{R}^2$, this would be asking whether the second vector lies 'clockwise' or 'counterclockwise' from the first, while in $\mathbb{R}^3$ the question is whether the vectors are 'left-handed' or 'right-handed'. A map that gives us this information is known as an ***orientation***. We will formalize this idea later, when we have introduced the necessary tools, those being vector spaces, their ***bases***, and ***determinant functions***.

In the complex case, we could dodge this issue because one of our vectors, call it $\vec{x}$, was always part of the x-axis, and thus we could obtain a canonical orientation by declaring that the angle between $\vec{x}$ and $\vec{y}$ was positive if $y_2 > 0$ and negative if $y_2 < 0$.

Exercise 1.12. *Prove the following theorems of classical Euclidean geometry:*

1. *The* **Law of Cosines***:*
$$\|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2 \cdot \cos\big(\measuredangle(\vec{x}, \vec{y})\big)$$

2. *The* **Pythagorean Theorem***:*
$$\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2} \implies \|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2$$

1.3 The Cross Product

The ***cross product*** is an operation $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$, such that:

1. $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
2. the length of $\vec{u} \times \vec{v}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$:

$$\left\|\vec{u} \times \vec{v}\right\|_2 = \left\|\vec{u}\right\|_2 \left\|\vec{v}\right\|_2 \sin\big(\measuredangle(\vec{u}, \vec{v})\big)$$

We also declare that our cross product should be ***anticommutative:*** given two vectors, we want to be able to change the sign of the cross product by switching the order in which we multiply:

$$\vec{v} \times \vec{w} := -\vec{w} \times \vec{v}$$

Notice how this immediately implies

$$\begin{aligned} & \vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) \\ \implies & \vec{v} \times \vec{v} = 0 \end{aligned}$$

Finally, notice that our geometric intuition immediately tells us that our product should be ***linear*** and ***distributive over vector addition***.

We can now derive an explicit formula for this operation on our own. First, note that the standard basis

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

consists of orthogonal vectors of length one, so we can start by declaring

$$\begin{aligned} \vec{e}_1 \times \vec{e}_2 &:= \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_3 &:= \vec{e}_1 \\ \vec{e}_3 \times \vec{e}_1 &:= \vec{e}_2 \end{aligned}$$

Note that once again, the direction in which these vectors point depends on a somewhat arbitrary choice of orientation - we could invert all of the directions and still end up with a valid definition.

These observations are already enough to calculate the cross product of two arbitrary vectors: Using distributivity, we get:

$$\begin{aligned}
\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} &= (v_1\vec{e}_1 + v_2\vec{e}_2 + v_3\vec{e}_3) \times (w_1\vec{e}_1 + w_2\vec{e}_2 + w_3\vec{e}_3) \\
&= v_1w_1(\vec{e}_1 \times \vec{e}_1) \\
&+ v_1w_2(\vec{e}_1 \times \vec{e}_2) \\
&+ v_1w_3(\vec{e}_1 \times \vec{e}_3) \\
\\
&+ v_2w_1(\vec{e}_2 \times \vec{e}_1) \\
&+ v_2w_2(\vec{e}_2 \times \vec{e}_2) \\
&+ v_2w_3(\vec{e}_2 \times \vec{e}_3) \\
\\
&+ v_3w_1(\vec{e}_3 \times \vec{e}_1) \\
&+ v_3w_2(\vec{e}_3 \times \vec{e}_2) \\
&+ v_3w_3(\vec{e}_3 \times \vec{e}_3)
\end{aligned}$$

To which we can apply anticommutativity and our formulas for multiplying the basis vectors to get:

$$\begin{aligned}
&\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \\
\\
&= v_1w_1 \cdot \vec{0} \\
&+ v_1w_2 \cdot \vec{e}_3 \\
&+ v_1w_3 \cdot (-\vec{e}_2) \\
\\
&+ v_2w_1 \cdot (-\vec{e}_3) \\
&+ v_2w_2 \cdot \vec{0} \\
&+ v_2w_3 \cdot \vec{e}_1 \\
\\
&+ v_3w_1 \cdot \vec{e}_2 \\
&+ v_3w_2 \cdot (-\vec{e}_1) \\
&+ v_3w_3 \cdot \vec{0} \\
\\
&= \begin{pmatrix} v_2w_3 - v_3w_2 \\ v_3w_1 - w_1v_3 \\ v_1w_2 - v_2w_1 \end{pmatrix}
\end{aligned}$$

Corollary 1.13: Cross Product

The cross product $\vec{v} \times \vec{w}$ of two vectors $\vec{v}, \vec{w} \in \mathbb{R}^3$ returns a vector whose direction is orthogonal to both $\vec{v}$ and $\vec{w}$ and whose length equals the area of the parallelogral spanned between them. Given coordinate representations of $\vec{v}$ and $\vec{w}$, the cross product can be calculated via the following formula:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$$

Proposition 1.14. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \times \vec{t} \right\rangle = \langle \vec{u}, \vec{w} \rangle \left\langle \vec{v}, \vec{t} \right\rangle - \left\langle \vec{u}, \vec{t} \right\rangle \langle \vec{v}, \vec{w} \rangle .$$

Proposition 1.15. *We have*

$$\langle \vec{u} \times \vec{v}, \vec{w} \rangle = \langle \vec{v} \times \vec{w}, \vec{u} \rangle = \langle \vec{w} \times \vec{u}, \vec{v} \rangle$$

This expression is known as the **box product** *of $\vec{u}, \vec{v}, \vec{w}$. Its absolute value corresponds to the volume of the parallelepiped spanned by the three vectors.*

Proposition 1.16. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} = \langle u, w \rangle \cdot v - \langle v, w \rangle \cdot u = w \times (v \times u).$$

This result is known as **Grassmann's Identity**.

Proposition 1.17. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} + (\vec{v} \times \vec{w}) \times \vec{u} + (\vec{w} \times \vec{u}) \times \vec{v} = 0$$

This result is known as **Jacobi's Identity.**

1.4 The Quaternions $\mathbb{H}$

Definition 1.18: Quaternions

The Quaternions $\mathbb{H}$ consist of the set $\mathbb{R}^4$, with the following special notation:

$$\begin{aligned} \vec{1} &:= (1,0,0,0) \\ i &:= (0,1,0,0) \\ j &:= (0,0,1,0) \\ k &:= (0,0,0,1) \end{aligned}$$

Such that in addition to scalar multiplication and vector addition, which work as usual, we have an additional vector multiplication operation $\cdot : \mathbb{R}^4 \times \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ that behaves as follows:

1. $\vec{1}$ is a neutral element of vector multiplication:

$$\begin{gathered} \forall x \in \mathbb{H} : \\ \vec{1} \cdot x = x \cdot \vec{1} = x \end{gathered}$$

2. Vector multiplication is associative:

$$\vec{x}(\vec{y}\vec{z}) = (\vec{x}\vec{y})\vec{z}$$

3. Multiplicative inverses exist:

$$\begin{gathered} \forall \vec{x} \in \mathbb{H}_{\neq 0} : \\ \exists \vec{x}^{-1} \in \mathbb{H} : \\ \vec{x} \cdot \vec{x}^{-1} = \vec{x}^{-1} \cdot \vec{x} = \vec{1} \end{gathered}$$

4. Vector multiplication distributes over vector addition, from both sides:

$$\begin{gathered} \forall \vec{x}, \vec{y}, \vec{z} \in \mathbb{H} : \\ \vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x}\vec{y} + \vec{x}\vec{z} \\ (\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x}\vec{z} + \vec{y}\vec{z} \end{gathered}$$

5. Vector multiplication is compatible with scalar multiplication:

$$\begin{gathered} \forall a, b \in \mathbb{R}, \vec{x}, \vec{y} \in \mathbb{H} : \\ a\vec{x} \cdot b\vec{y} = (ab)(\vec{x}\vec{y}) \end{gathered}$$

6. The unit vectors fulfill the ***Quaternion multiplication formula***:

$$i^2 = j^2 = k^2 = ijk = -\vec{1}$$

Apart from the Quaternion multiplication formula, which is the defining feature of the Quaternions, all of our other demands together state that $\mathbb{H}$ should be an ***associative real division algebra over*** $\mathbb{R}^4$.

One very important consequence of this definition is that vectors of the form $\vec{a} := (a, 0, 0, 0)$ behave exactly like real numbers - let $\vec{x} \in \mathbb{H}$, then we have:

$$\vec{a}\vec{x} = (a \cdot \vec{1}) \cdot \vec{x} = a \cdot (\vec{1} \cdot \vec{x}) = a\vec{x}$$

For this reason, it is common to write $\vec{1}$ simply as 1, and to then proceed as we did for the complex numbers and write:

$$(a, b, c, d) = a + bi + cj + dk$$

We can now already get most of the way to deriving multiplication formula for quaternions, by simply applying distributivity:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i + (v_2 u_2) i^2 + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji + (v_3 u_3) j^2 + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj + (v_4 u_4) k^2
\end{aligned}$$

Now, using $i^2 + j^2 + k^2 = ijk = -1$, we can simplify slightly:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i + (v_2 u_2) i^2 + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji + (v_3 u_3) j^2 + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj + (v_4 u_4) k^2 \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4)
\end{aligned}$$

All that's left is to figure out what the products of i, j and k should be.

We can derive ij and ji as follows:

1. Start with $ijk = -1$.
2. Multiply k from the right to get $ijk^2 = -k$, i.e. $-ij = -k$, i.e. $ij = k$.
3. Multiply ij from the left to get $ijij = ijk = -1$.
4. Multiply i from the left and j from the right to get $i(ijij)j = -ij$.
5. Simplify the left side to get $i(ijij)j = (ii)(ji)(jj) = ji$, i.e:

$$ij = -ji = k$$

Exercise 1.19. *Show that we also have $jk = -kj = i$ and $ki = -ik = j$.*

This means that Quaternion multiplication isn't commutative!

Our final multiplication table for $1, i, j$ and k is thus:

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Exercise 1.20. *Verify that this multiplication table doesn't violate associativity.*

Entering these new results into our partial multiplication formula, we get:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\quad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\quad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik \\
&\quad + (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk \\
&\quad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4) \\
&\quad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\quad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) k - (v_2 u_4) j \\
&\quad + (v_3 u_1) j - (v_3 u_2) k - (v_3 u_3) + (v_3 u_4) i \\
&\quad + (v_4 u_1) k + (v_4 u_2) j - (v_4 u_3) i - (v_4 u_4)
\end{aligned}$$

which we can finally simplify to get:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\quad = (v_1 u_1 - v_2 u_2 - v_3 u_3 - v_4 u_4) \\
&\quad + (v_1 u_2 + v_2 u_1 + v_3 u_4 - v_4 u_3) i \\
&\quad + (v_1 u_3 - v_2 u_4 + v_3 u_1 + v_4 u_2) j \\
&\quad + (v_1 u_4 + v_2 u_3 - v_3 u_2 + v_4 u_1) k
\end{aligned}$$

Now, this is obviously a long and unreasonably complicated formula - is there any way to think of it differently? The complex numbers had a very nice and intuitive geometric definition, maybe the quaternions do too?

Well, the key trick here is to split the underlying set $\mathbb{R}^4$ into $\mathbb{R} \times \mathbb{R}^3$, thinking of each Quaternion as consisting of a ***real part*** $a \in \mathbb{R}$ and an ***imaginary part*** $\vec{v} \in \mathbb{R}^3$. Our multiplication formula now reads:

$$\begin{aligned}
&(a, \vec{v}) \cdot (b, \vec{u}) \\
&= (ab - v_1 u_1 - v_2 u_2 - v_3 u_3) \\
&+ (a u_1 + b v_1 + v_2 u_3 - v_3 u_2) i \\
&+ (a u_2 - v_1 u_3 + b v_2 + v_3 u_1) j \\
&+ (a u_3 + v_1 u_2 - v_2 u_1 + b v_3) k
\end{aligned}$$

which you might notice consists exclusively of familiar vector operations:

1. $ab - v_1u_1 - v_2u_2 - v_3u_3 = ab - \left\langle \vec{v}, \vec{u} \right\rangle$

2. $\begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} = a\vec{u}, \begin{pmatrix} bv_1 \\ bv_2 \\ bv_3 \end{pmatrix} = b\vec{v}$

3. $\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_1u_3 - v_3u_1 \\ v_1u_2 - v_2u_1 \end{pmatrix} = \vec{v} \times \vec{u}$

Putting everything together, we can now write out our multiplication formula in a much nicer form:

$$(a, \vec{v}) \cdot (b, \vec{u}) = (ab - \left\langle \vec{v}, \vec{u} \right\rangle, a\vec{u} + b\vec{v} + \vec{v} \times \vec{u}),$$

Exercise 1.21. *Verify that quaternion multiplication is associative and distributes over vector addition.*

Exercise 1.22. *Let $p = (a, \vec{v}) \in \mathbb{H}$. Then $pq = qp$ holds for all $q \in \mathbb{H}$ if and only if $\vec{v} \neq 0$.*

Similar to complex numbers, we can define ***quaternion conjugation***:

Definition 1.23: Quaternion Conjugation

Just like for $\mathbb{C}$, we define the ***conjugate*** of a quaternion $(a, \vec{v})$ to be the quaternion

$$\overline{(a, \vec{v})} = (a, -\vec{v})$$

An immediate application of quaternion conjugation is finding a formula for the inverse of a quaternion, which ones again ends up working out very similarly to the complex case - We have

$$\begin{aligned} \overline{(a, \vec{v})} \cdot (a, \vec{v}) &= (a, -\vec{v}) \cdot (a, \vec{v}) \\ &= (a^2 - \left\langle -\vec{v}, \vec{v} \right\rangle, a\vec{v} - a\vec{v} + (-\vec{v} \times \vec{v})) \\ &= \left(a^2 + \left\|\vec{v}\right\|^2, \vec{0}\right) \in \mathbb{R} \end{aligned}$$

And thus we can simply divide by this value to get our inverse formula:

Corollary 1.24: Inverse Quaternion

Let $(a, \vec{v}) \in \mathbb{H}_{\neq 0}$. Then $(a, \vec{v})$ has a multiplicative inverse, which is given by:

$$(a, \vec{v})^{-1} = \frac{(a, -\vec{v})}{a^2 + \|v\|^2}.$$

Definition 1.25. Let $q = (a, \vec{v}) \in \mathbb{H}$. Then

$$|q| := \sqrt{\overline{q} \cdot q} = a^2 + v_1^2 + v_2^2 + v_3^2 = \|q\|_2$$

Note how much this once again resembles both the real case $|x| = \sqrt{x^2} = \|x\|_2$ and the complex case $|z| = \sqrt{\overline{z}z} = \|z\|_2$.

1.5 Quaternions and $\mathbb{R}^3$

Theorem 1.26: Quaternions encode rotations

Let $\|\vec{v}\| = 1$ and

$$q = (\cos\varphi, \vec{v}\sin\varphi) \in \mathbb{H}$$

Then $qw\overline{q} \in \mathbb{R}^3$ is the vector obtained by rotating w around the axis $\{\lambda\vec{v} : \lambda \in \mathbb{R}\}$ by the angle 2φ.

Since 2π corresponds to a full rotation, φ and $\varphi + \pi$ describe the same rotation. Therefore, there are two different quaternions q and $-q$ associated with each rotation - in topological terms, we say that the quaternions form a ***double cover*** of the space of all rotations.

This phenomenon is actually closely related to the theory of ***spin*** in quantum physics. All known quantum particles that constitute ordinary matter, including protons, neutrons, electrons, neutrinos, and quarks, are ***spin-$\frac{1}{2}$***, meaning that they exhibit a curious behavior where a rotation of 360° leaves them in a different state than they were before, from which they return to their original state only after being rotated by another 360° around the same axis. This lets us mathematically model particles as quaternions, such that a 360° rotation transforms a particle from a state $q \in \mathbb{H}$ to the "rotationally equivalent" state $-q \in \mathbb{H}$.

Corollary 1.27. *For every isometry $F : \mathbb{R}^3 \to \mathbb{R}^3$, there exist $u \in \mathbb{R}^3$ and $q \in \mathbb{H}$ such that we have either*

1. $F(w) = u + qw\overline{q}$, *if F preserves orientation, or*
2. $F(w) = u + q\overline{w}\overline{q}$, *if F doesn't preserve orientation.*

This representation turns out to be a lot more computationally efficient than other representations of the same rotation. Other, more intuitively "trivial" representations of rotation also run into issues such as ***gimbal lock***, where they may get "stuck" in certain orientations.

This makes quaternions surprisingly practical in many applied contexts, including game development and animation[1], robotics, and three-dimensional image processing.

[1] For an introduction to quaternions in game development, I highly recommend this excellent talk by Freya Holmer.

Chapter 2

Linear Maps

2.1 Modules and Vector Spaces

Definition 2.1: Modules

Let $(R, +, \cdot)$ be a Ring. Then a (left) ***R-Module*** consists of an abelian Group $(M, +)$ equipped with a ***scalar multiplication*** $. : R \times M \to M$, i.e. scalars are elements of R and "vectors" are elements of M, such that:

1. Ring multiplication associates with scalar multiplication:
$$(r \cdot s).m = r.(s.m)$$
2. Scalar multiplication distributes over module addition:
$$r.(m + n) = r.m + r.n$$
3. Scalar multiplication distributes over scalar addition:
$$(r + s).m = r.m + s.m$$

If R is unital, then we say that M is unital if the identity of ring multiplication is the identity of scalar multiplication:

$$1.m = m$$

A ***right R-module*** is defined similarly, but with a scalar multiplication $. : M \times R \to M$, which is applied from the right instead of from the left.

Exercise 2.2. *Show that the distinction between right R-modules and left R-modules is only relevant if R is noncommutative, i.e. that any left R-module M over a commutative ring can be turned into a right R-module by defining $m.r = r.m$ and vice versa.*

To keep this text in line with familiar conventions of vector multiplication in $\mathbb{R}^n$, whenever we talk about a module without explicitly specifiying whether scalar multiplication is applied from the right or from the left, it is implied that it is applied from the left.

Exercise 2.3. *Let M be a module over R. Let $m \in M$ and $r \in R$. show that:*

1. $0_R.m = 0_M$
2. $r.0_M = 0_M$
3. $(-r).m = r.(-m) = -(r.m)$

Exercise 2.4. *Verify that:*

1. *Every ring R forms a module over itself, where scalar multiplication is given by ring multiplication.*
2. *$\mathbb{R}^n$ forms a module over $\mathbb{R}$, with scalar multiplication defined the usual way. More generally, given any ring R, the cartesian product R^n forms an R-module where scalar multiplication is given by ring multiplication applied componentwise.*
3. *Any abelian group A can be turned into a non-unital module over any ring R by equipping it with the trivial scalar multiplication*

$$r.a = 0_A$$

Definition 2.5: Vector Spaces

A module over a field (or skew field) $\mathbb{k}$ is known as a $\mathbb{k}$-***vector space***.

Definition 2.6: Linear Maps

A ***module homomorphism*** is a structure-preserving map $F : M \to N$ between two modules over the same ring R, i.e. given $m_1, m_2 \in M$ and $r \in R$, we have that:

1. F is ***additive***, i.e. it is compatible with module addition:

$$F(m_1 +_M m_2) = F(m_1) +_N F(m_2)$$

2. F is ***homogenous***, i.e. it is compatible with scalar multiplication:

$$F(r._M m) = r._N F(m)$$

Module homomorphisms are also known as ***linear maps***.

As the name suggests, the main goal of linear algebra is to generalize the concept of a ***linear map***, and to figure out properties and applications of these maps.

Exercise 2.7. *Show that any linear map $F : M \to N$ fulfills*

$$F(0_M) = 0_N$$

Exercise 2.8. *Show that any linear map $F : R \to R$ from a unital ring into itself can be represented as multiplication by a constant, i.e. $F(r) = a.r = a \cdot_R r$ for some $a \in R$.*

Exercise 2.9.

1. *Show that complex conjugation is a linear map $\mathbb{C} \to \mathbb{C}$ if we view $\mathbb{C}$ as a module over $\mathbb{R}$.*
2. *Show that complex conjugation is not linear if we view $\mathbb{C}$ as a module over $\mathbb{C}$.*

2.2 Subspaces and Quotient Spaces

2.3 Bases and Dimension

Definition 2.10: Linear Combinations

Let M be an R-module, let $(m_i)_{i\in I} \subseteq M$ and $(r_i)_{i\in I} \subseteq R$. Then a ***linear combination*** is a sum of the form

$$\sum_{i\in I} r_i.m_i$$

Definition 2.11: Vector Span

Let M be an R-module, let $S = (m_i)_{i\in I} \subseteq M$. Then the span of S is the set $\langle S\rangle$ of all linear combinations over elements of S, i.e.

$$\langle S\rangle = \left\{ \sum_{i\in J\subseteq I} r_i.m_i \;\middle|\; (r_i)_{i\in I} \subseteq R, m_i \in S \right\}$$

For example, the span of one vector $v \in \mathbb{R}^3$ is the line through the origin on which v lies. The span of two orthogonal vectors $v, w \in \mathbb{R}^3$ is the plane through the origin on which v and w lie.

2.4 Matrices

2.5 Systems of Linear Equations

2.6 The Method of Least Squares

Chapter 3

Affine Spaces

Affine spaces are structures similar to vector spaces, but where the notions of "points" and "directions" are separate.

Definition 3.1: Affine Space

A $\Bbbk$-***affine space*** is a tuple $(A, \vec{A})$, consisting of a set A of ***points*** and a $\Bbbk$-vector space $\vec{A}$ of ***translation vectors***, alongside a translation map $+ : A \times \vec{A} \to A$, such that:

1. Translation is associative:

$$\forall a \in A, \vec{v}, \vec{w} \in \vec{A} :$$
$$(a + \vec{v}) + \vec{w} = a + (\vec{v} + \vec{w})$$

2. For any pair of points, there exists a unique vector translating one point onto the other:

$$\forall a, b \in P_A :$$
$$\exists! \vec{v} \in \vec{A} : a + \vec{v} = b$$

The dimension of an affine space is simply the dimension of its associated vector space.

Any vector space V can be canonically turned into an affine space by taking V as both the set of points and the set of vectors. On the flipside, we can turn any affine space $(A, \vec{A})$ into a vector space by restricting our attention to translations of one "base point" $b \in A$, and then identifying any point $p \in A$ with the vector $\vec{v} \in \vec{A}$ translating b onto p.

For most practical purposes, it is therefore sufficient to work with a vector space as your base space for whatever math you are doing, but the non-canonicity of having to choose an arbitrary base point $b \in A$ to turn an affine space into a vector space means that some people prefer affine spaces for philosophical reasons - after all, what is the "base point" of our physical universe?

The distinction becomes a lot more important when considering affine subspaces, which are essentialy vector subspaces translated by arbitrary points:

Definition 3.2: Affine subspaces

Let $(A, \vec{A})$ be an affine space. Then an affine subspace of $(A, \vec{A})$ is a tuple $(B, \vec{B})$ such that:

1. $B \subseteq A$.
2. $\vec{B}$ is a vector subspace of $\vec{A}$.
3. $\vec{B}$ is the set of translations between points in B, i.e. for any $b \in B$, $\vec{v} \in \vec{A}$, we have $b + \vec{v} \in B$ if and only if $\vec{v} \in \vec{B}$.

Thus, an affine subspace can always be written as

$$B = b + \vec{B} := \left\{ b + \vec{v} \,\middle|\, \vec{v} \in \vec{B} \subseteq \vec{A} \right\}$$

for an arbitrary base point $b \in B$.

An affine subspace of dimension 1 is known as a ***line***. An affine subspace of dimension 2 is known as a ***plane***. An affine subspace whose dimension is one lower than the space it resides in is known as an ***hyperplane***.

Definition 3.3. Two affine subspaces $(A, \vec{A})$, $(B, \vec{B})$ are called ***parallel*** if $\vec{A} = \vec{B}$.

Definition 3.4: Affine Maps

Let A, B be $\mathbb{k}$-affine spaces. Then a map $F : A \longrightarrow B$ is called ***affine*** if there exists a linear map $\vec{F} : \vec{A} \longrightarrow \vec{B}$ such that

$$\forall a \in P_a, \vec{v} \in \vec{A} : F(a + \vec{v}) = F(a) + \vec{F}(\vec{v})$$

As previously mentioned, if our affine space is a vector space, we can identify any point in P_A with the vector translating the origin onto that point, which lets us write:

$$F(\vec{v}) := F(0 + \vec{v}) = F(0) + \vec{F}(\vec{v}) := \vec{F}(\vec{v}) + \vec{w}$$

thus, an affine map between vector spaces is simply a linear map composed with a translation.

Theorem 3.5: Intercept Theorem

Let $v_1, v_2 \in \mathbb{R}$ be linearly independent. Let L_1 be an affine line through v_2 and v_1. Then if we have $\alpha, \beta \in \mathbb{R}_{\neq 0}$, and L_2 is a parallel line through αv_2 and βv_1, we must have $\alpha = \beta$.

Proof. Since L_1 and L_2 are parallel, their direction vectors must point in the same direction, which means there must be a $\mu \in \mathbb{R}_{\neq 0}$ such that

$$\begin{aligned}
&\mu(v_2 - v_1) = \mu(\alpha v_2 - \beta v_1) \\
\implies &\mu v_2 - \mu v_1 = \alpha v_2 - \beta v_1 \\
\implies &\mu v_2 - \alpha v_2 = \mu v_1 - \beta v_1 \\
\implies &(\mu - \alpha) v_2 = (\mu - \beta) v_1
\end{aligned}$$

Since v_2 and v_1 are linearly independent, this implies $\mu - \alpha = \mu - \beta = 0$, i.e. $\alpha = \beta = \mu$. □

Theorem 3.6. *An injective map $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$ is linear if and only if it maps $\vec{0}$ to $\vec{0}$ and lines to lines.*

Proof. $\Rightarrow$ Let Φ be linear. Let $g := \left\{\vec{p} + \lambda \cdot \vec{v}\right\}$ be a line. Then we have

$$\begin{aligned}
\Phi(g) &= \left\{\Phi(\vec{p} + \lambda \cdot \vec{v})\right\} \\
&= \left\{\Phi(\vec{p}) + \lambda \cdot \Phi(\vec{v})\right\},
\end{aligned}$$

i.e. $\Phi(g)$ is also a line.

$\Leftarrow$ Since Φ maps $\vec{0}$ to $\vec{0}$ and lines to lines, it must map lines through the origin to lines through the origin. Let

$$\begin{aligned}
\Phi(e_1) &:= v_1 \\
\Phi[\lambda e_1] &:= L_1 = \{\lambda v_1\} \\
\Phi(e_2) &:= v_2 \\
\Phi[\lambda e_2] &:= L_2 = \{\lambda v_2\}
\end{aligned}$$

Since Φ is injective, L_1 and L_2 can only intersect at $\{0\}$, and thus v_1 and v_2 must be linearly independent.

Now, let's take a closer look at images of individual points. Since $\Phi[\lambda e_1] = \{\lambda v_1\}$ and $\Phi[\lambda e_2] = \{\lambda v_2\}$, we must have

$$\begin{aligned}
\Phi[\lambda e_1] &= \varphi_1(\lambda) v_1 \\
\Phi[\lambda e_2] &= \varphi_2(\lambda) v_2
\end{aligned}$$

for some $\varphi_1, \varphi_2 : \mathbb{R} \to \mathbb{R}$.

Now, notice that the line through λe_1 and λe_2 is parallel to the line through e_1 and e_2. Since Φ is injective, it must preserve parallelism, and so the line through v_1 and v_2 must be parallel to the line through $\varphi_1(\lambda)v_1$ and $\varphi_2(\lambda)v_2$. The intercept theorem now gives us $\varphi_1(\lambda) = \varphi_2(\lambda) = \mu$. We thus write $\varphi(\lambda) := \varphi_1(\lambda) = \varphi_2(\lambda)$.

Since Φ is injective, φ must be as well, since otherwise we would have some v with $\Phi(\lambda_1 v) = \Phi(\lambda_2 v)$ despite $\lambda_1 v \neq \lambda_2 v$. We have also have $\varphi(0) = 0$ and $\varphi(1) = 1$, since $\Phi(0) = 0$ and $\Phi(e_1) = v_1$.

Now, we want to show that $\Phi(v + w) = \Phi(v) + \Phi(w)$ holds for linearly independent v, w. Since Φ preserves parallelism, it must map parallelograms to parallelograms. Therefore, the parallogram spanned by v and w must be mapped to the parallogram spanned by $\Phi(v)$ and $\Phi(w)$. This means that the upper corner $v + w$ gets mapped to the upper corner $\Phi(v) + \Phi(w)$ which gives us $\Phi(v + w) = \Phi(v) + \Phi(w)$ as desired.

Putting our insights together, we now know that

$$\begin{aligned}\Phi(\lambda_1 e_1 + \lambda_2 e_2) &= \Phi(\lambda_1 e_1) + \Phi(\lambda_2 e_2)\\ &= \varphi(\lambda_1)\Phi(e_1) + \varphi(\lambda_2)\Phi(e_2)\end{aligned}$$

it only remains to be shown that φ is the identity. For this, consider that we have:

$$\begin{aligned}(\varphi(a) + \varphi(b))\Phi(e_1) &= \varphi(a)\Phi(e_1) + \varphi(b)\Phi(e_1)\\ &= \Phi(ae_1 + be_1\\ &= \Phi((a + b)e_1)\\ &= \varphi(a + b)\Phi(e_1),\end{aligned}$$

which gives us $\varphi(a) + \varphi(b) = \varphi(a + b)$. For $a \neq 0$, the line through e_1 and ae_2 must point in the same direction as the line through be_1 and bae_2, so the line through $\Phi(e_1)$ and $\varphi(a)\Phi(e_2)$ must be parallel to the line through $\varphi(b)\Phi(e_1)$ and $\varphi(ab)\Phi(e_2)$, from which the intercept theorem gives us

$$\begin{aligned}\frac{\varphi(ab)}{\varphi(a)} &= \varphi(b)\\ \implies \varphi(ab) &= \varphi(a)\varphi(b)\end{aligned}$$

Since we have $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi(a + b) = \varphi(a) + \varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, our map $\varphi : \mathbb{R} \to \mathbb{R}$ must be a field homomorphism, of which the identity is the only one. Thus, Φ is linear.

□

Exercise 3.7. *An injective map between real affine spaces of dimension ≥ 2 is affine if and only if it maps lines to lines.*

Chapter 4

Determinants

4.1 Defining Volume

4.2 Determinants of Endomorphisms

4.3 Orientation

Chapter 5

Eigenvectors and Eigenvalues

Definition 5.1: Eigenvalues and Eigenvectors

Let k be a field, $F : V \to V$ an Endomorphism, and $\lambda \in k$. Then the ***λ-eigenspace*** of F is the set

$$\begin{aligned}\mathrm{Eig}(F, \lambda) &:= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda\mathrm{id}_V)\end{aligned}$$

If $F(\vec{v}) = \lambda\vec{v}$, with $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an ***eigenvector*** of F with ***eigenvalue*** λ.

Note that we exclude $\vec{0}$ from being an eigenvector, since $F(\vec{0}) = \lambda\vec{0}$ trivially holds for every endomorphism. Thus every λ would be an eigenvalue, which would make the definition pointless. We do, however, allow 0 as an eigenvalue - the 0-eigenspace of F is simply its kernel. Sometimes it ends up being useful to rephrase our definition of eigenspaces in terms of kernels:

$$\begin{aligned}\mathrm{Eig}(F, \lambda) &= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : F(\vec{v}) - \lambda\vec{v} = 0\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda\mathrm{id}_V)\end{aligned}$$

This shows that $\mathrm{Eig}(F, \lambda)$ is a subspace of V - by definition, these subspaces must then be the maximal subspaces that are closed under application of F, i.e. $F(\mathrm{Eig}(F, \lambda)) = \mathrm{Eig}(F, \lambda)$.

Exercise 5.2. *In an informal, intuition-based way:*

1. *Find the real eigenvalues and eigenspaces of a reflection of $\mathbb{R}^2$ along an axis $\left\{\lambda\vec{v} : \lambda \in \mathbb{R}\right\}$*

2. *Find a rotation of $\mathbb{R}^2$ that has no real eigenvalues. Find two different rotations of $\mathbb{R}^2$ that do have real eigenvalues. Show that any rotation of $\mathbb{R}^2$ has complex eigenvalues.*

3. *Find the eigenvectors of the derivative map $\frac{d}{dx}$*

Our characterisation $\mathrm{Eig}(F, \lambda) = \ker(F - \lambda \mathrm{id}_V)$ means that, if F is finite-dimensional, we can find all eigenvectors of F with a given eigenvalue λ by solving $(F - \lambda \mathrm{id}_V)\vec{v} = 0$, which is simply a systen of linear equations. Finding the correct eigenvalues ends up being more tricky - in the section after this one, we will work towards proving the following key result:

Proposition 5.3. *The eigenvalues of F are given by the zeroes of the* **characteristic polynomial** $\chi_F(\lambda) := \det(F - \lambda \mathrm{id}_V)$.

However, before we do so, we want to familiarize ourselves with some of the more basic properties of eigenvectors and eigenvalues.

One important application is that finding eigenvectors lets us diagonalize endomorphisms, greatly easing computation afterwards:

Theorem 5.4. *Let V be a vector space over a field k. Let $B = (b_1, \ldots, b_n)$ be a basis of V, and let F be an endomorphism represented in B by a matrix A. Then:*

1. *A basis vector b_i is an eigenvector of F iff. the j-th column of A has the form λe_j.*
2. *F is diagonalizable iff. there exists a basis of V consisting of eigenvectors of F.*

Theorem 5.5. *Let V be a vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then eigenvectors of F with different eigenvalues are linearly independent.*

Corollary 5.6. *Let V be an n-dimensional vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then:*

1. *F has no more than n distinct eigenvalues,*
2. *If $(\lambda_i)_{i \in I}$ are mutually distinct eigenvalues of F, the sum*
$$\sum_{i \in I} \mathrm{Eig}(F, \lambda_i) \subseteq V$$
is direct.
3. *F is diagonalizable if and only if V is a direct sum of eigenspaces of F.*
4. *If F has n distinct eigenvalues, it is diagonalizable.*

Corollary 5.7. *Let V, W be n-dimensional vector spaces over a field k. Let $F : V \longrightarrow V$, $G : W \longrightarrow W$ be endomorphisms. Then there exists a unique endomorphism $P : V \longrightarrow W$ such that $P \circ F = G \circ P$ if and only if F and G have the same eigenvalues.*

5.1 Characteristic Polynomials

Theorem 5.8: Characterstic Polynomial

Let V be a finite-dimensional vector space over a field $\mathbb{k}$. Let $F : V \longrightarrow V$ be an endomorphism. Then $\lambda \in \mathbb{k}$ is an eigenvalue of F if and only if

$$\chi_F(\lambda) = \det(\lambda \cdot \mathrm{id}_V - F) = 0_{\mathbb{k}}$$

We call χ_F the ***characteristic polynomial of F***.

Proof. We know that $\lambda \in \Bbbk$ is an Eigenvalue of F if and only if

$$\ker(F - \lambda \mathrm{id}_V) \neq \{0_V\},$$

i.e. if and only if $F - \lambda \mathrm{id}_V$ isn't injective.

We also know that $F - \lambda \mathrm{id}_V$ isn't injective if and only if it isn't surjective, i.e. if and only if it isn't invertible, i.e. if and only if

$$\det(\lambda.\mathrm{id}_V - F) = \chi_F(\lambda) = 0_{\Bbbk}.$$

□

Exercise 5.9. *Show that every endomorphism $F : V \to V$ of a finite-dimensional complex vector space V has at least one eigenvalue.*

Exercise 5.10. *Show that the constant term of a characteristic polynomial $\chi_F(\lambda)$ is given by $(-1)^n \cdot (\det F)$.*

(...)

5.2 Algebras and the Cayley-Hamilton Theorem

Theorem 5.11: Evaluation map

Let R be a commutative unital ring, and let A be a commutative unital algebra over R. Then, for every $a \in A$, there exists a unique algebra homomorphism $\mathrm{ev}_a : R[X] \to A$ such that:

1. $\mathrm{ev}_a(r) = 1_A.r$ for all constant polynomials $r \in R[X]$,
2. $\mathrm{ev}_A(X) = X$.

We call ev_a the ***evaluation map of $R[X]$ at a***.

Informally, this means that we can take elements of an algebra A over R and plug them into polynomials that were originally only defined on R, and we won't run into issues. In particular, we can plug matrices into polynomials over an underlying ring! Since ev_a is unique, we will generally just write $P(a) := \mathrm{ev}_a(P)$ from now on.

Theorem 5.12: Cayley-Hamilton

Let R be a commutative unital ring. Let $A \in R^{n \times n}$. Then $\chi_A(A) = 0$.

5.3 Minimal Polynomials

Theorem 5.13: Minimal polynomial

et F be an endomorphism of a vector space V over a field $\Bbbk$. Then there exists a unique polynomial $\mu_F \in \Bbbk[X]$ such that:

1. μ_F is either normalized or equal to $0_{\Bbbk[X]}$
2. μ_F is a divisor of every $P \in \Bbbk[X]$ with $P(F) = 0_{\mathrm{End}_{\Bbbk}(V)}$

We call μ_F the ***minimal polynomial of F***.

Chapter 6

Normal Forms of Endomorphisms

6.1 Decomposing Vector Spaces using Polynomials

Exercise 6.1. *Let G be an abelian group. Show that G becomes a $\mathbb{Z}$-module when equipped with the operation* $n.g = \underbrace{g + \ldots + g}_{n \text{ times}}$.

Exercise 6.2. *Let V be a vector space over a field $\Bbbk$. Show that V becomes an* $\mathrm{End}(V)$*-Module when equipped with the operation $F.v = F(v)$.*

Theorem 6.3. *Any $\Bbbk[X]$-module is the same thing as a $\Bbbk$-vector space V equipped with a specified endomorphism F, and vice versa.*

Proof.

1. Let V be a $\Bbbk$-vector space. We know from the previous exercise that V is an $\mathrm{End}(V)$-module. Thus, the polynomial evaluation map $\mathrm{ev}_F : \Bbbk[X] \to \mathrm{End}(V)$ turns V into $\Bbbk$-module via

$$P.v = \mathrm{ev}_F(P)(v) = P(F)(v)$$

2. Now let M be a $\Bbbk[X]$-module. Then M becomes a $\Bbbk$-module, i.e. a $\Bbbk$-vector space, if we restrict scalar multiplication to only accept constant polynomials. Then it can easily be seen that $G(v) = X.v$ is a linear transformation.

Now, note that if we start with a vector space (V, F), then use our construction to get a module M, and then turn it back into a vector space, we will obtain our original F again, since

$$G(v) = X._M v = X(F)(v) = F(v)$$

Conversely, if we start with a module M, then turn it into a vector space (V, F), and then back into a module M, we reobtain our original scalar multiplication, since

$$P._V v = P(G)(v) = P(X._M)(v) = P._M v$$

Therefore, the two directions of our construction are inverse to each other, i.e. we actually have a bijection. □

Thus, given a $\Bbbk[X]$-module M, we can write $M = (V, F)$

Exercise 6.4. *Let $M = (V, F)$ and $N = (W, G)$ be $\Bbbk[X]$-Modules. Then a $\Bbbk[X]$-linear map $H : M \to N$ is $\Bbbk$-linear and additionally fulfills $H \circ F = G \circ H$, and any $\Bbbk$-linear map H fulfilling these properties is $\Bbbk[X]$-linear.*

Let $M = (V, F)$ be a $\Bbbk[X]$-Module. Then, by definition, a subspace $U \subseteq V$ is a $\Bbbk[X]$-submodule of M if and only if U is closed under scalar multiplication with elements of $\Bbbk[X]$.

Exercise 6.5. *Show that U is closed under scalar multiplication with elements of $\Bbbk[X]$ if and only if $Xu \in U$, which itself holds if and only if $F(u) \in U$ holds for all $u \in U$, i.e. U is **F-invariant**.*

Exercise 6.6.

1. *Show that the scalar multiplication map $P. = P(F)$ is linear.*
2. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are submodules of M.*
3. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are F-invariant.*

Theorem 6.7: Structure theorem for modules over a Euclidean ring

Let M be a module over a euclidean domain R. Let $a = bc \in R$ such that b and c share no divisors and $a.m = 0$ for all $m \in M$. Then we have:

1. $\ker(b.) = \operatorname{im}(c.)$,
2. $\operatorname{im}(b.) = \ker(c.)$,
3. $M = \ker(b.) \oplus \ker(c.)$.

Proof.

1. Let $c.m \in \operatorname{im}(c.)$ for $m \in M$. Then we have $b.(c.m) = a.m = 0$, i.e. $\operatorname{im}(c.) \subseteq \ker(b.)$. Now, applying the Euclidean algorithm gives us $1 = k_1 b + k_2 c$. If we now let $m \in \ker(b.)$, we get:

$$\begin{aligned} m &= 1m \\ &= (k_1 b + k_2 c)m \\ &= k_1 \underbrace{(bm)}_{=0} + k_2(cm) \\ &= c(k_2 m) \in \operatorname{im}(c.) \end{aligned}$$

   Thus, we also have $\ker(b.) \subseteq \operatorname{im}(c.)$, i.e. $\ker(b.) = \operatorname{im}(c.)$.

2. $\ker(c.) = \operatorname{im}(b.)$ follows from the same argument by swapping the roles of b and c.

3. Now, let $m \in M$. Then, applying our identity from before, we get

$$\begin{aligned} m &= (k_1 b + k_2 c)m \\ &= b(k_1 m) + c(k_2 m) \\ &\in \operatorname{im}(b.) + \operatorname{im}(c.) \\ &= \ker(b.) + \ker(c.), \end{aligned}$$

which gives us $M = \ker(b.) + \ker(c.)$ as desired.

If we have $m \in \ker(b.) \cap \ker(c.)$, we get

$$m = k_1(b(m)) + k_2(c(m)) = 0,$$

so $\ker(b.) \cap \ker(c.) = \{0\}$, so our sum is direct.

□

Corollary 6.8. *Let G be an abelian group. Let $n = mk \in \mathbb{Z}$ such that m and k share no divisors and $n \cdot g = 0$ for all $g \in G$. Then we have*

$$G = \ker(m\cdot) \oplus \ker(k\cdot).$$

Corollary 6.9: Chinese Remainder Theorem

Corollary 6.10. *Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P = QR \in \Bbbk[X]$ such that Q and R share no divisors and $P.v = P(F)(v) = 0$ for all $v \in V$. Then we have*

$$V = \ker(Q(F)) \oplus \ker(R(F)).$$

Corollary 6.11: P-Generalized Eigenspace Decomposition

Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P \in \Bbbk[X]$ be a normed polynomial and let $P = P_1^{n_1} \cdot \ldots \cdot P_k^{n_k}$ be its prime factorization. Then we have

$$V = \bigoplus_{i=1}^{k} \ker(P_i^{n_i}(F))$$

6.2 The Jordan Normal Form

Corollary 6.12: Generalized Eigenspace Decomposition

Let V be a vector space over a field $\mathbb{k}$ and $F : V \to V$ be an Endomorphism. Assume that its minimal polynomial μ_F splits into linear factors over $\mathbb{k}$, i.e.

$$\mu_F(x) = \prod_{i=1}^{k}(x - \lambda_i)^{n_i}$$

with $\lambda_i \in \mathbb{k}$. Then we have

$$V = \bigoplus_{i=1}^{k} \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

We call $\ker((F - \lambda_i \mathrm{id}_V)^{n_i})$ the ***generalized λ_i-Eigenspace of F*** and denote it $H(F, \lambda_i)$ (from the German "Hauptraum").

Exercise 6.13. *Show that $H(F, \lambda_i)$ contains all eigenvectors of F with eigenvalue λ_i.*

Exercise 6.14. *Show that $H(F, \lambda_i)$ is an F-invariant subspace of V, i.e. a subspace such that $F(H(F, \lambda_i)) \subseteq H(F, \lambda_i)$*

Exercise 6.15. *Let $V = V_1 \oplus V_2$ be a decomposition of V into F-invariant subspaces. Let B_1 be a basis of V_1, and let B_2 be a basis of V_2. Show that:*

1. *$B = B_1 \cup B_2$ is a basis of V*
2. *If the restriction $F|_{V_1}$ of F to V_1 is represented in basis B_1 as a matrix A_1, and the restriction $F|_{V_2}$ of F to V_2 is represented in basis B_2 as a matrix A_2, then F is represented in basis B as*

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Exercise 6.16. *Let $F : V \to W$ be a linear map. Let A be a basis of* $\ker F$*, and let B be a tuple of vectors such that $F(B)$ is a basis of* $\mathrm{im}\, F$*. Then $A \cup B$ is a basis of V.*

We now want to make use of this to construct a Jordan basis B of a nilpotent Endomorphism N:

Corollary 6.17. *Let $N : V \to V$ be a nilpotent endomorphism. Then there exists a basis B of V such that:*

1. *$B \cup \{0\}$ is F-invariant, i.e. if $b \in B$, then we have $F(b) \in B$ or $F(b) = 0$.*
2. *For every $b \in B$, there exists at most one element $b' \in B$ with $F(b') = b$.*

We call such a basis a **Jordan basis***.*

> *Proof.* We prove this theorem in three steps: We construct a basis B, then show it is F-invariant, then show that each element of B has at most one preimage in B.

1. We construct our basis by induction over $n = \dim V$.

   If $n = 0$, then the empty set is a basis (since a sum over the empty set equals the additive identity, i.e. the zero vector) which (vacuously) fulfills our desired properties.

   Now, since the kernel of N is non-trivial, we have $\dim \operatorname{im} N < \dim V$. Therefore, by induction, we can assume we already have a Jordan basis A of im N. We split this basis A into two parts:

   (a) $A_0 := \{a \in A \mid N(a) = 0\}$,

   (b) $A_{\neq 0} := \{a \in A \mid N(a) \neq 0\}$.

   A_0 is by definition a set of linearly independent vectors contained in $\ker A$. By the Steinitz exchange lemma, we can add a set C of Linearly independent vectors such that $A_0 \cup C$ is a basis of $\ker A$.

   $A_{\neq}0$ is the set of all A such that $N(A_{\neq 0}) \subseteq A$. We add a set D containing preimages of all the remaining elements of A, which must exist since $A \subseteq \operatorname{im} N$. By construction, we now have $F(A_{\neq 0} \cup D) = A$.

   Putting everything together, $A_0 \cup C$ is a basis of $\ker\ N$, and $F(A_{\neq 0} \cup D) = A$, which is a basis of im N. Therefore, by exercise 6.16, $B = A \cup C \cup D$ is a basis of V.

2. Now, B must be F-invariant since A is F-invariant by definition, C is contained in the kernel of F and thus $F(B) = \{0\}$, and D consists of preimages of A, i.e. $F(D) = A \subseteq B$.

3. Finally, the preimages of A in B are contained in either $A_{\neq 0}$ or D, which are disjoint by construction.

□

Corollary 6.18. *Let $F : V \to V$ be a matrix whose characteristic polynomial has $n = \dim(V)$ roots. Then there exists a basis B of V such that ${}_B F_B$ is a block diagonal matrix consisting of Jordan blocks.*

Proof. Generalized Eigenspace decomposition 6.2 shows us that V is the direct sum of the generalized Eigenspaces of F. We know by exercise 6.14 that the generalized Eigenspaces of F are F-invariant. Therefore, exercise 6.15 tells us that if we find a basis B_i for each generalized Eigenspace $H(F, \lambda_i)$ such that $F|_{H(F,\lambda_i)}$ is represented by a matrix J_i in Jordan normal form, and we combine our bases into a basis $B = \bigcup_{i=1}^{n} B_i$ of V as a whole, then the matrix representation of F in basis B will be a block diagonal matrix consisting of these J_i, and thus F will also be in Jordan normal form.

Now, let F_i be the restriction of $(F - \lambda_i \mathrm{id}_V)$ to $H(F, \lambda_i)$. By definition of the generalized Eigenspace, we have

$$H(F, \lambda_i) = \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

therefore, by construction, F_i is nilpotent. Therefore, by theorem **??**, there exists a basis B_i such that ${}_{B_i}F_{i\,B_i}$ is in Jordan normal form, with zeroes on the diagonal. But then, since we have

$$F|_{H(F,\lambda_i)} = F_i + \lambda \mathrm{id}_{H(F,\lambda_i)}$$

the matrix representation of $F|_{H(F,\lambda_i)}$ is also in Jordan form, since we simply have to add λ to the diagonal of ${}_{B_i}F_{i\,B_i}$.

Therefore, the restriction to F to any of its generalized Eigenspaces is in Jordan normal form in basis B_i. As discussed at the beginning of this proof, this means F is in Jordan normal form in basis $B = \bigcup_{i=1}^{n} B_i$. □

Exercise 6.19. *Let F be an endomorphism, and let J be its Jordan normal form. Then:*

1. *The algebraic multiplicity of each eigenvalue of F corresponds to the number of times that eigenvalue appears on the diagonal of J.*
2. *The geometric multiplicity of any eigenvalue λ of F corresponds to the number of λ-Jordan blocks with that eigenvalue in J.*
3. *The exponent of $(X - \lambda)$ in the minimal polynomial of F corresponds to the size the of largest λ-Jordan block in J.*

6.3 The Weierstrass Normal Form

Chapter 7

Inner Product Spaces

7.1 Inner Products

Definition 7.1. Let $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let V and W be vector spaces over $\Bbbk$. Then a map $\varphi : V \to W$ is called ($\Bbbk$-) ***semilinear*** or ***antilinear***, if we have

1. ***Addivity:*** $\forall u, v \in V : \varphi(u + v = \varphi(u) + \varphi(v)$
2. ***Antihomogeneity:*** $\forall v \in V, \lambda \in \Bbbk : \varphi(\lambda.v) = \overline{\lambda}.\varphi(v)$.

Notice that for $\Bbbk = \mathbb{R}$, there is no difference between linearity and antilinearity, since complex conjugation leaves real numbers unchanged.

Definition 7.2. Let $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let V be a $\Bbbk$-vector space. Then a map $s : V \times V \to \Bbbk$ is called a ***sesquilinear form*** if it is antilinear in the first component and linear in the second component,i.e. the map $s_w : v \to s(v, w)$ is antilinear and the map $s_v : w \to s(v, w)$ is linear.

Exercise 7.3. *Let A be a matrix with entries in $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Show that the map*

$$s_A(v, w) := v^* A w$$

is sesquilinear. (Recall that v^* stands for the conjugated transpose of v.)

The latin prefix "sesqui-" means "one and a half" - a sesquilinear is a map that is linear in one component, and "sort of linear", i.e. "one-half-linear" in the other. Some sources require that a sesquilinear form is linear in the first component and antilinear in the second.

Definition 7.4. A sesquilinear form $s : V \times V \to \Bbbk$ is called a ***Hermitian form*** if it is ***conjugate symmetric***, i.e. $s(v, w) = \overline{s(w, v)}$.

Once again, notice that for $\Bbbk = \mathbb{R}$, this just means $s(v, w) = s(w, v)$.

Exercise 7.5. *Show that a matrix is Hermitian (i.e. fulfills $A = A^*$) if and only if its associated sesquilinear form $s_A(v, w) = v^* A w$ is Hermitian.*

Definition 7.6. A sesquilinear form $s : V \times V \to \Bbbk$ is called ***positive semidefinite*** if $s(v, v) \geq 0$ for all $v \in V$. It is called ***positive definite*** if it additionally fulfills $s(v, v) = 0$ if and only if $v = \vec{0}$.

Positive definiteness is the primary reason we are working with sesquilinear forms instead of bilinear forms - we need to be able to compare $s(v, v)$ to 0, and there is no way to order all of $\mathbb{C}$ that is compatible with its field structure. However, if we have a sesquilinear Hermitian map, we get $s(v, v) = \overline{s(v, v)}$, i.e. $s(v, v) \in \mathbb{R}$, which means we can start comparing things again.

Definition 7.7. A ***scalar product*** or ***inner product*** on a vector space V over $\mathbb{R}$ or $\mathbb{C}$ is a positive definite Hermitian form. We call a real vector space equipped with a scalar product ***Euclidean***, and a complex vector space equipped with a scalar product ***unitary***.

Definition 7.8. The standard scalar product on $\mathbb{C}^n$ is given by

$$\langle v, w \rangle_{\mathbb{C}} = v^* w$$

Exercise 7.9. *Show that $\langle v, w \rangle_{\mathbb{C}}$ is actually a scalar product.*

Exercise 7.10. *Let $V = C^\infty([0,1], \mathbb{R})$ be the room of infinitely differentiable functions $[0,1] \to \mathbb{R}$. Show that the $\boldsymbol{L^2}$ scalar product*

$$\langle f, g \rangle_{L^2} = \int_0^1 f(t) g(t)\, dt$$

is actually a scalar product.

Definition 7.11. Let V be a vector space over $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then a ***Norm*** is a map $\|\cdot\| : V \to \mathbb{k}$ fulfilling the following three axioms:

1. ***Positive definiteness:*** $\|v\| \geq 0$ and $\|v\| = 0$ iff. $v = \vec{0}$
2. ***Homogeneity:*** $\|\lambda . v\| = |\lambda| . \|v\|$
3. ***Triangle inequality:*** $\|v + w\| \leq \|v\| + \|w\|$

A norm is a way of assigning lengths to vectors. Once lengths of vectors are defined, one can easily define distances between vectors too:

Exercise 7.12. *Let $\|\cdot\|$ be a norm on a vector space V over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Show that $d(v, w) := \|v - w\|$ is a metric on V.*

Theorem 7.13: Cauchy-Schwartz Inequality

Let s be a scalar product on a vector space V over a field $\mathbb{k}$. Then we have

$$|\langle v, w \rangle| \leq \sqrt{\langle v, v \rangle} \cdot \sqrt{\langle w, w \rangle}$$

One important property of scalar products is that they induce norms. Notice that the standard norm on $\mathbb{R}^n$ is just the square root of the standard scalar product on $\mathbb{R}^n$. This property generalizes to all scalar products:

Exercise 7.14. *Let $s : V \times V \to \mathbb{k}$ be a scalar product on a vector space V over a field $\mathbb{k}$. Let $\|v\|_s := \sqrt{s(v, v)}$. Show that:*

1. *$\|\cdot\|_s$ is a norm on V.*
2. *We have the* **parallelogram identity**

$$\|v + w\|_s^2 + \|v - w\|_s^2 = 2\|v\|_s^2 + 2\|w\|_s^2$$

3. *If $\mathbb{k} = \mathbb{R}$, we have the* **real polarisation formula**

$$s(v, w) = \frac{1}{4}\Big(\|v + w\|_s^2 - \|v - w\|_s^2\Big)$$

4. *If $\Bbbk = \mathbb{C}$, we have the* **complex polarisation formula**

$$s(v, w) = \frac{1}{4}\Big(\|v + w\|_s^2 - \|v - w\|_s^2 - i\|v + i.w\|_s^2 + i\|v - i.w\|_s^2\Big)$$

Exercise 7.15. *Find a norm on a vector space that cannot be constructed as the square root of a scalar product.*

Theorem 7.16. *Let V be a vector space over $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let $\|\cdot\|$ be a norm on V fulfilling the parallelogram identity. Then there exists a corresponding scalar product s such that $\|v\| = \sqrt{s(v, v)}$*

7.2 Orthogonality

Definition 7.17. Let (V, s) be a vector space with scalar product over a field $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Then we call two vectors v, w ***orthogonal*** iff.

$$s(v, w) = 0$$

Exercise 7.18. *Let $B = (v_1, \dots, v_k)$ be a tuple of orthogonal vectors not containing the zero vector. Then B is linearly independent.*

Definition 7.19. A basis consisting of pairwise orthogonal vectors is called an ***orthogonal basis***. If the basis vectors are also normalized, i.e. have length 1, we call the basis an ***orthonormal basis***.

Exercise 7.20. *Let (V, s) be a vector space with scalar product. Show that $B = (b_1, \dots, b_n) \subset V$ is an orthonormal basis iff.*

$$s(b_i, b_j) = \delta_{ij}$$

(Recall that δ_{ij} is defined as the function that is 1 if $i = j$ and 0 otherwise).

Definition 7.21. Let (V, s) be a vector space with scalar product over a field $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let $U \subseteq V$ be a subspace. Let $B_U := \{b_1, \dots, b_m\}$ be a basis of U. Then the **orthogonal projection of V onto U** is the map

$$\begin{aligned} p : V &\to U \\ v &\mapsto \sum_{i=1}^{m} g(b_i, v).b_i \end{aligned}$$

Exercise 7.22. 1. *Show that the orthogonal projection of V onto U leaves vectors in u unchanged.*

2. *Show that for each $v \in V$ and $u \in U$, u is orthogonal to $v - p(v)$.*

3. *Show that for each $v \in V$, $p(v)$ is the unique point in U for which $\|v - p(v)\|_s$ is minimal.*

Theorem 7.23. *Let (V, s) be a vector space with scalar product over a field $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let $(v_1, \dots, v_n)$ be a basis of V. Then there are unique vectors $(b_1, \dots, b_n) \subset V$ such that for all $p \in (1, \dots, n)$, we have:*

1. *$(b_1, \dots, b_p)$ is an s-orthonormal Basis of $\langle v_1, \dots, v_p \rangle \subseteq V$*

2. *$s(b_p, v_p) \in \mathbb{R}_{>0}$*

These vectors b_p can be constructed inductively via $b_1 = \frac{v_1}{\|v_1\|}$ followed by projection onto the already constructed basis vectors, i.e.

$$
\begin{aligned}
w_p &= v_p - \sum_{q=1}^{p-1} g(b_q, v_p).b_q \\
b_p &= \frac{w_p}{\|w_p\|}
\end{aligned}
$$

Corollary 7.24. *Every vector space with scalar product has an orthonormal basis.*

7.3 Matrix Representations of Inner Products

Definition 7.25. Let s be a sesquilinear form on a finite dimensional vector space Vover a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $B = (b_1, \ldots, b_n)$ be a basis of V. Then the ***Gram matrix of B by g*** is defined as the matrix with entries

$$G_{ij} = s(b_i, b_j) \in \mathbb{k}^{n\times n}$$

Exercise 7.26. *Show that the Gram matrix of the standard basis of $\mathbb{R}^n$ by the scalar product is the identity matrix.*

Definition 7.27. A matrix $A \in \mathbb{k}^{n\times n}$ is called ***positive (semi-)definite*** if $s_A(v, w) = v^*Aw$ is positive (semi-)definite.

Exercise 7.28. *Let s be a sesquilinear form on a finite-dimensional vector space V over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let G be its Gram matrix. Then we have $s(v, w) = v^*Gy$ in basis B. In particular, s is a scalar product if and only if its Gram matrix is Hermitian and positive definite.*

Theorem 7.29. *Let $A \in \mathbb{k}^{n\times n}$, where $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then the following are equivalent:*

1. *A is hermitian and psitive definite.*
2. **Cholesky Decomposition:** *There exists an upper triangular Matrix B with positive real diagonal entries such that $A = B^*B$.*
3. *There exists an invertible Matrix B such that $A = B^*B$.*
4. **Sylvester-Hurwitz Criterion:** *A is Hermitian and and all of its square upper left submatrices A_r, known as its* **leading principal minors**, *have strictly positive determinant.*

7.4 Dual Spaces

Definition 7.30: Dual Space

Let V be a vector space over a field $\mathbb{k}$. Then we define $V^* = \mathrm{Hom}_{\mathbb{k}}(V, \mathbb{k})$. Elements of V^*, i.e. linear maps $V \to \mathbb{k}$, are known as ***linear forms***.

Theorem 7.31: Dual Basis

iven a basis $B = (b_1, \dots, b_n)$ of a vector space V, the set of linear forms $E = (\varepsilon_1, \dots, \varepsilon_n)$ given by

$$\varepsilon_p(e_q) = \delta_{pq}$$

is a basis of V^*, known as the ***dual basis*** of B.

Exercise 7.32. *Let $C^n(\mathbb{k}, \mathbb{k})$, $n \in 0, 1, \dots, \infty$ be the set of n-times continuously differentiable functions $\mathbb{k} \to \mathbb{k}$. Then for any $x_0 \in \mathbb{k}$, the map*

$$\cdot(x_0) : f \to f^{(n)}(x_0)$$

is a linear form on $C^n(\mathbb{k}, \mathbb{k})$ (where $f^{(n)}$ denotes the n-th derivative of f).

Exercise 7.33. *The map mapping an integrable function f to $\int_0^1 f(x)\,dx$ is a linear form. More generally, for any integrable $g : [0, 1] \to \mathbb{k}$, the map taking f to $\int_0^1 \overline{g(x)} \cdot f(x)\,dx$ is a linear form.*

Theorem 7.34. *Let (V, g) be a $\mathbb{k}$-vector space with scalar product. Then g induces an injective antilinear map $g_1 : V \to V^*$ via*

$$g_1(v) = g(v, \cdot)$$

A linear form $\alpha \in V^*$ is called **representable** *by g if there exists a $v \in V$ such that $\alpha = g(v, \cdot)$.*

Definition 7.35. Let V be a $\mathbb{k}$-vector space. Then a ***antilinear form*** is an antilinear forms $\gamma : V \to \mathbb{k}$. The ***antidual space*** $\overline{V}^*$ of V is the space of antilinear maps $V \to \mathbb{k}$.

Theorem 7.36. *Let (V, g) be a finite-dimensional vector space over a field $\mathbb{k}$. Then the maps $g : V \to V^*$ and $\overline{g} : V \to V^*$ are bijective.*

Corollary 7.37. *Let (V, g) be a finite-dimensional vector space. Then all linear and antilinear forms on V are representable by g.*

Definition 7.38. Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called ***adjointable*** if there exists a map $G : W \to V$ such that

$$g(G(w), v) = h(w, F(v)$$

for all $v \in V$ and $w \in W$. Such a map G is known as the ***adjoint of F***, and is written $G = F^*$. If $F = F^*$, we call F ***self-adjoint***.

Proposition 7.39. *If it exists, the adjoint of a given map is unique.*

Proposition 7.40. *Adjoint maps are linear.*

Proposition 7.41. *If G is the adjoint of F, then F is the adjoint of G, i.e. we have $(F^*)^* = F$.*

Proposition 7.42. *If A is a matrix representation of F by two orthonormal bases, then the conjugate transpose A^* is a matrix representation of F^*.*

Definition 7.43. Let V and W be $\Bbbk$-vector spaces, with $\Bbbk$ a field. Let $F : V \to W$ be a linear map. Then the ***dual*** of F is the map

$$\begin{aligned} F^\vee : W^* &\to V^* \\ \beta &\mapsto \beta \circ F \end{aligned}$$

i.e.

$$F^\vee(\beta)(\vec{v}) = \beta(F(\vec{v}))$$

for $\beta \in W^*, \vec{v} \in V$.

We can construct a dual of any linear map this way, while linear maps involving infinite-dimensional vector spaces are generally not adjoinable.

Exercise 7.44. *Verify the following properties of the dual map:*

1. $(\mathrm{id}_V)^\vee = \mathrm{id}_{V^*}$
2. *Let $F : V \to W$ and $G : U \to V$ be linear, then $(F \circ G)^\vee = G^\vee \circ F^\vee$*
3. *The dual of a linear map is linear.*
4. *If $F : V \to W$ is given by $F(\vec{v}) = A \cdot \vec{v}$ for some matrix A, then $F^\vee(\beta) = \beta \cdot A$, where elements $\beta \in W^*$ are represented as row vectors.*

Theorem 7.45. *Let $U \subseteq V$ be vector spaces. Let $\iota : U \to V$ be the inclusion map. Let $q : V \to V/U$ be the quotient map. Then $(V/U)^* \cong \mathrm{im}\ q^\vee = \ker\ \iota^\vee$.*

Proof. Since q maps V to V/U, we have $q^\vee : (V/U)^* \to \mathrm{im}\ q^* \subseteq V^*$. Since q is linear, so is $q^\vee$. By definition, any map surjectively maps onto its image. Therefore, all we need to show to obtain $(V/U)^* \cong \mathrm{im}\ q^\vee$ is injectivity.

So, let $\beta \in \ker\ q^\vee \subseteq (V/U)^*$, i.e. let $q^\vee(\beta) = \beta(q)$ be the zero map. Since q is surjective, $\beta(q(v)) = 0$ for all $v \in V$ already implies $\beta(w) = 0$ for all $w \in V/U$, so β must have already been the zero map.

Finally, we want to show $\mathrm{im}\ q^\vee = \ker\ \iota^\vee$.

First off, recall that $\iota^\vee : V^* \to U^*$. By the universal property of the quotient map, for every $\beta \in \ker \iota^\vee \subseteq V^*$, there exists $\overline{\beta} : V/U \to \Bbbk$ (i.e. $\overline{\beta} \in (V/U)^*$) with $\beta = \overline{\beta} \circ q = \overline{\beta}(q) \in \mathrm{im}\ q^\vee$, so we have $\ker \iota^\vee \subseteq \mathrm{im}\ q^\vee$.

Conversely, for any $\beta \in (V/U)^*$ and $v \in (V/U)$, we have

$$q^\vee(\beta)(v) = \beta(q(v)) = \beta(0) = 0,$$

so we also have $\mathrm{im} q^\vee \subseteq \ker \iota^\vee$. □

Theorem 7.46

Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Let $h_1 : W \to W^*$ and $g_1 : V \to V^*$ be the antilinear maps induced by g and h, i.e.

$$h_1(\vec{w}) = h(\vec{w}, \cdot)$$
$$g_1(\vec{v}) = g(\vec{v}, \cdot)$$

Let $F : V \to W$ be linear. Then $g_1 \circ F^* = F^\vee \circ h_1$.

Proof. Let $v \in V, w \in W$, then we have:

$$\begin{aligned} g_1(F^*(w))(v) &= g(F^*(w), v) \\ &= h(w, F(v)) \\ &= h_1(w)(F(v)) \\ &= F^\vee(h_1(w))(v) \end{aligned}$$

□

Corollary 7.47. *If V is finite-dimensional, $F : V \to W$ is adjointable.*

Proof. If V is finite-dimensional, g_1 is invertible. We get $F^* = g_1^{-1} \circ F^\vee \circ h_1$. □

Definition 7.48. Let V and W be $\mathbb{k}$-vector spaces. Let $F : V \to W$ be linear, and let $s : W \times W \to \mathbb{k}$ be a sesquilinear form. Then we define the ***pullback of s by F*** to be the map $F^*s : V \times V \to \mathbb{k}$ given by

$$(F^*s)(u, v) = s(F(u), F(v))$$

Exercise 7.49. *Verify that:*

1. *F^*s is a sesquilinear form.*
2. *If s is Hermitian, F^*s is Hermitian.*
3. *If s is positive semidefinite, F^*s is positive semidefinite.*
4. *If s is positive definite and F is injective, F^*s positive definite.*
5. *If we fix bases of V and W such that F is represented by a matrix A and s is represented by a Gram matrix G, then F^*s is represented by A^*GA.*
6. *Given $F : V \to W$ antilinear, we can achieve the same results by defining $F^*s(u, v) = s(F(v), F(u))$.*

Chapter 8

Normal Endomorphisms

Definition 8.1

Let (V, g) be a $\Bbbk$-vector space with scalar product. Let $F : V \to V$ be an adjointable endomorphism. Then we call F ***normal*** if $F^* \circ F = F \circ F^*$.

Exercise 8.2. 1. *Show that self-adjoint maps are normal.*

2. *A linear map with $F^* = -F$ is called* **skew**. *Show that skew maps are normal.*

3. *A linear map represented in a given basis by a matric of the form*

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal.

Definition 8.3

Let (V, g) and (W, h) be $\Bbbk$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called an ***isometric embedding*** if $g(v, w) = h(F(v), F(w))$ for all $v, w \in V$.

Exercise 8.4. *Show that:*

1. *Isometric embeddings preserve lengths and distances.*

2. *Isometric embeddings preserve angles.*

3. *Isometric embeddings are injective.*

4. *If orthonormal bases are chosen for V and W, then the matrix representation A of F fulfills $A^*A = E_n$.*

5. *If* $\dim V = \dim W$, *F is invertible with $F^{-1} = F^*$. An invertible isometric embedding is called an* **isometry**.

6. *Isometries are normal.*

Theorem 8.5: Fundamental Theorem of Normal Endomorphisms

Let (V, g) be a finite-dimensional complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) consisting of eigenvectors of F if and only if F is normal.

Note that this only works for complex numbers, not for reals:

Exercise 8.6. *Recall that any endomorphism represented by a matrix of the form*

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal. Show that, if $b \neq 0$, A has no real eigenvalues.

Corollary 8.7: A Normal Form for Complex Normal Endomorphisms

Let (V, g) be a finite-dimensional real vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $b > 0$ if and only if F is normal. If such a representation exists, it is unique up to permutation of the blocks.

Corollary 8.8: Spectral Theorem for Self-Adjoint Maps

Let (V, g) be a finite dimensional real or complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of V in which F is represented by a diagonal matrix with real entries if and only if F is self-adjoint.

Exercise 8.9. *Derive an analogous spectral theorem for skew maps.*

Corollary 8.10. *Let (V, g) be a finite-dimensional real or complex vector space with scalar product. Let s be a sesquilinear form on V. Then there exists a g-orthonormal Basis of V in which s is represented by a diagonal matrix with real entries if and only s is hermitian.*

Corollary 8.11: Singular Value Decomposition

Let (V, g) and (W, h) be finite-dimensional real or complex vector spaces with scalar products. Let $F : V \to W$ be linear. Then there exist orthonormal bases of V and W in which F is represented by a diagonal matrix with uniquely determined real entries a_n such that $a_1 \geq a_2 \geq \dots$ and $a_n = 0$ if $n \geq \operatorname{rk} F$. These entries are known as the ***singular values*** of F.

Singular value decomposition is fundamental to the JPG compression algorithm, which represents an image by a matrix, then calculates the singular value decomposition of that matrix, and then sets all of the singular values below a given threshold to zero, leading to less data needing to be stored while still maintaining the singular values that 'noticably contribute' to the resulting image.

8.1 Linear Isometries

Corollary 8.12: Normal Form for Real Isometries

Let (V, g) be a finite-dimensional real vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $a^2 + b^2 = 1$.

Corollary 8.13: Normal Form for Complex Isometries

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a diagonal matrix whose entries have absolute value 1.

Corollary 8.14

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then V can be decomposed into a direct sum of subspaces U of dimension one or two, such that F acts on the one-dimensional subspaces as the identity or reflection, and on the two-dimensional subspaces as a rotation around the origin.

Exercise 8.15. *A linear isometry is orientation-preserving if and only if the number of invariant subspaces on which it acts as a reflection is even.*

8.2 Affine Isometries

Theorem 8.16. *Let $(A, \vec{A}, s))$, $(B, \vec{B}, g)$ be finite-dimensional normed affine spaces of the same dimension over vector spaces with scalar products. Then there exists an affine isometry $F : A \to B$.*

Theorem 8.17. *Let $(A, \vec{A}, g)$ be a finite-dimensional normed $\Bbbk$-affine space over a vector space with scalar product. Let $F : A \to A$ be an affine sometry. Then there exists* *a point $o \in A$ and a vector $x \in \operatorname{Eig}(\vec{F}, 1)$ such that*

$$F(a) = o + x + \vec{F}(a - o)$$

x is uniquely determined by A, while o is an arbitrary point chosen from an affine-F-invariant subspace $(B, \operatorname{Eig}(\vec{F}, 1)) \subseteq (A, \vec{A})$.

Definition 8.18. We call $(B, \operatorname{Eig}(\vec{F}, 1))$ the ***axis set*** of F, and call lines of the form

$$\{b + \lambda.x \mid \lambda \in \Bbbk\} \subseteq B$$

with $b \in B$ the ***axes*** of F.

Our theorem thus states that any affine isometry can be represented as a linear map composed with a translation along the axis set of F.

Proof. We write $F(a) = \vec{F}(a) + y$. Let $U = \operatorname{Eig}(\vec{F}, 1)$. Then U and $U^\perp$ are both invariant under $\vec{F}$, and $\vec{F}|_U = \mathrm{id}_U$.

We write $y = (x, z) \in U \oplus U^\perp$. The map $\mathrm{id}_{U^\perp} - \vec{F}|_{U^\perp}$ is invertible, since $U^\perp$ doesn't contain any eigenvectors of $\vec{F}$. We can thus find a $q \in U^\perp$ with $q - \vec{F}(q) = z$.

We can then choose any $p \in U$ and set $o = (p, q)$. For any other $v = (u, w) \in V$, we then get

$$\begin{aligned}
F(a) &:= F(o + v) \\
&= y + \vec{F}(o + v) \\
&= (x, z) + \vec{F}((p, q) + (u, w)) \\
&= (x, z) + \vec{F}((p + u, q + w)) \\
&= (x + \vec{F}(p) + \vec{F}(u), z + \vec{F}(q) + \vec{F}(w)) \\
&= (x + p + u, z + \vec{F}(q) + \vec{F}(w)) \\
&= (x + p + u, (q - \vec{F}(q)) + \vec{F}(q) + \vec{F}(w)) \\
&= (x + p + u, q + \vec{F}(w)) \\
&= (p, q) + (x, 0) + \vec{F}(u, w) \\
&= o + (x, 0) + \vec{F}(v), \\
&= o + x + \vec{F}(a - o),
\end{aligned}$$

which is our desired representation.

Now, assume we chose an alternate origin $o' = o + (p', q')$ such that

$$F(a) := F(o' + v) = o' + x' + \vec{F}(v)$$

for $v = (u, w)$. Then we get:

$$
\begin{aligned}
x' &= F(o' + v) - o' - \vec{F}(v) \\
&= F(o + (p', q') + v) - o - (p', q') - \vec{F}(v) \\
&= (o + x + \vec{F}((p', q') + v)) - o - (p', q') - \vec{F}(v) \\
&= (o + x + \vec{F}(p', q') + \vec{F}(v) - o - (p', q') - \vec{F}(v) \\
&= x + \vec{F}(p', q') - (p', q') \\
&= x + \vec{(}p', \vec{F}(q')) - (p', q') \\
&= (x, \vec{F}(q') - q')
\end{aligned}
$$

Now, applying $\vec{F}$ to x and x', we get:

$$
\begin{aligned}
\vec{F}(x) &= F(x) - (x, z) \\
&= F((x, 0)) - (x, z) \\
&= o + (x, 0) + \vec{F}((x, 0) - o) - (x, z) \\
&= o + (x, 0) + (x, 0) - \vec{F}(o) - (x, z) \\
&= o + (x, 0) + (x, 0) - (p, q - z) - (x, z) \\
&= (p, q) + (x, 0) + (x, 0) - (p, q) + (0, z) - (x, z) \\
&= (x, 0) = x
\end{aligned}
$$

and thus

$$
\begin{aligned}
\vec{F}(x') &= (\vec{F}(x), \vec{F}(\vec{F}(q')) - \vec{F}(q')) \\
&= (x, \vec{F}(\vec{F}(q')) - \vec{F}(q'))
\end{aligned}
$$

therefore, we get $\vec{F}(x') = x'$ if and only if

$$\vec{F}(\vec{F}(q')) - \vec{F}(q') = \vec{F}(q') - q',$$

i.e.

$$\vec{F}(\vec{F}(q')) - 2\vec{F}(q') + q' = 0,$$

i.e.

$$(\mathrm{id}_W - \vec{F}|_W)(\mathrm{id}_W - \vec{F}|_W(q')) = 0$$

Since $\mathrm{id}_W - \vec{F}|_W$ is invertible by construction, this holds iff. $q' = 0$, i.e. if $x' = (x, \vec{F}(q') - q') = x$ and $o' = o + (p', 0) = o + \lambda x \in B$. $\square$

Chapter 9

Revisiting Synthetic Geometry

9.1 Incidence Geometries

Definition 9.1: Incidence Geometries

An ***incidence geometry*** is a set X equipped with a set $G \subseteq \mathcal{P}(X)$ of subsets of X known as ***lines***, such that:

1. Each line contains at least two points,
2. For every pair of points, there exists exactly one line containing both.

Instead of "x is contained in $g \in G$", we often say "x lies on $g \in G$".

Exercise 9.2. *Let R be a division ring. Then $X := R^n$ can be turned into an incidence geometry by defining lines to be all sets of the form*

$$g = \{p + \lambda v : \lambda \in R\},$$

where $p, v \in R^n$.

Moving forward, this will be the canonical choice of incidence geometry on R^n.

Definition 9.3. A set of points lying on the same line is called ***colinear***.

Definition 9.4. A set of lines containing a common point is called ***confluent***.

Definition 9.5. A set of three non-colinear points is known as a ***triangle***.

Definition 9.6: Betweenness

A ***betweenness relation*** on an incidence geometry (X, G) is a subset $Z \subseteq X^3$ such that the following properties hold:

1. ***Induced Orderings on Lines:*** On every line $g \in G$, there exists an ordering $\leq$ such that $(x, y, z) \in Z$ if and only if $x \leq y \leq z$ or $x \geq y \geq z$.

2. ***Pasch's Axiom:*** Let $p, q, r \in X$. Let

$$[p, q] := \{x \in X : (p, x, q) \in Z\}$$

denote the ***line segment between p and q***. Then every line $g \in G$ either intersects none of the line segments $[p, q], [q, r], [r, p]$, or at least two.

An incidence geometry equipped with a betweenness relation is known as an ***ordered incidence geometry***.

Exercise 9.7. *Let K be an ordered field. Then:*

1. *The set of all tuples (x, y, z), where $x, y, z \in K^n$ fulfill $y = \lambda x + \mu z$ with $\lambda, \mu \in [0, 1] \subset K$ and $\lambda + \mu = 1$, gives a betweenness relation on K^n.*

2. *There exists an affine isomorphism $\varphi : K \cong (g \in G)$ between any line in G and K.*

3. *The ordering on K can be recovered from the betweenness relation on K^n by taking the unique ordering $\leq$ on $g \in G$ that fulfills $\varphi(0) \leq \varphi(1)$.*

Moving forward, this will be the canonical choice of betweenness relation on K^n.

Definition 9.8. Given an ordered incidence geometry (X, G, Z), a ***ray*** in X is a set $A \subseteq X$ of the form

$$A = \{x \in g \mid x \leq p\},$$

where $g \in G$ is a line and $p \in g$ is a point on that line.

Definition 9.9: Congruence Groups

Given an ordered incidence geometry (X, G, Z), a ***congruence group*** of (X, G, Z) is a group K of automorphism of (X, G, Z) such that every pair of rays in $A, B \subseteq X$ has exactly two elements $h, k \in K$ such that $h(A) = k(A) = B$.

Intuitively, one of these two elements will be a rotation and the other a rotation composed with reflection along an axis.

Exercise 9.10. *Let K be an ordered field. Let s be the standard scalar product in K^2. Then the set of s-invariant affine automorphisms of K^2 forms a congruence group of K^2.*

Moving forward, this will be the canonical choice of congruence group on K^2.

Definition 9.11. We say that an ordered incidence geometry has the ***least upper bound property*** if every nonempty subset $A \subset g \in G$ that is bounded from above has a least upper bound.

Definition 9.12. We call an ordered incidence geometry (X, G, Z) with congruence group K ***near-euclidean*** if (X, G, Z) fulfills the supremum property and G is nonempty.

Definition 9.13. We say that an incidence geometry (X, G) fulfills the ***parallel postulate*** iff. for every line $g \in G$ and every point $p \notin G$, there exists at most one line $h \in G$ such that $p \in h$ and $h \cap g = \emptyset$.

Exercise 9.14. *Let $K \subsetneq \mathbb{R}$ be a proper subfield of $\mathbb{R}$ containing square roots of every positive element. Then K^2 fulfills the parallel postulate.*

The primary goal of this chapter will be proving the following result:

Theorem 9.15: Classification of near-euclidean geometries

Let (X, G, Z) be a near-euclidean incidence geometry. Then:

1. If (X, G, Z) fulfills the parallel postulate, it is isomorphic to $\mathbb{R}^2$.
2. If (X, G, Z) doesn't fulfill the perallel postulate, it is isomorphic to the ***hyperbolic plane*** H.

9.2 Projection Maps

Definition 9.16. Let V be a vector space. Then a ***linear hyperplane*** is a vector subspace of W of codimension 1, i.e.

$$\operatorname{codim}(W \subseteq V) := \dim(V/W) = 1$$

Equivalently, $W \neq V$ such that V is generated by W and any vector $v \in V \setminus V$. If V is finite-dimensional, we have $\dim(V/W) = \dim V - \dim W$, and thus a linear hyperplane is a subspace W whose dimension is one lower than the dimension of V. An ***affine Hyperplane*** is defined entirely analogously as an affine subspace of codimension one.

Exercise 9.17.

1. *In any affine space, the intersection of an affine line with another affine subspace is either empty, a single point, or the entire line.*
2. *For any affine hyperplane H and any line g with directional vector $\vec{v} \notin \vec{H}$, the intersection of g and H consists of a single point.*

Definition 9.18. Given an affine space E, an affine hyperplane $H \subset E$, and a vector $\vec{v} \in \vec{E} \setminus \vec{H}$, the ***parallel projection of E onto H in direction $\vec{v}$*** is the map

$$\pi = \pi_{\vec{v}} = \pi_{\vec{v},H} : E \to H$$

assigning to each point e the intersection of the affine line $(e + \langle \vec{v} \rangle)$ with H.

Definition 9.19. Given an affine space E, an affine hyperplane $H \subset E$, and a point $q \in E \setminus H$, the ***central projection of E onto H centered at q*** is the map

$$\zeta = \zeta_q = \zeta_{q,H} : E \setminus (q + \vec{H}) \to H$$

assigning to each point $e \in E$ the intersection of the line though e and q with H.

Exercise 9.20. *The image of a line g under central projection is either:*

1. *A line,*
2. *a pointed line, i.e. a line with a single missing point (the* **vanishing point***),*
3. *a single point,*
4. *or the empty set.*

9.3 Projectivization

Definition 9.21: Projectivization

Let $\Bbbk$ be a field. Let V be a $\Bbbk$-vector space. Then the set of all lines through the origin in V is called the the ***projectivization of V*** and denoted by $\mathbb{P}V$.

In the finite-dimensional case, we have $\dim \mathbb{P}V = \dim V - 1$. It is therefore common to write

$$\mathbb{P}^n\Bbbk := \mathbb{P}(\Bbbk^{n+1})$$

Beware, this means $\mathbb{P}\Bbbk \neq \mathbb{P}^1\Bbbk$! The set $\mathbb{P}^1\Bbbk$ is called the ***projective line over*** $\Bbbk$, the set $\mathbb{P}^2\Bbbk$ is called the ***projective plane over*** $\Bbbk$.

Definition 9.22. Given a vector space V, we call a subset $g \subseteq \mathbb{P}V$ a ***projective line*** or $\mathbb{P}$***-line*** if there exists a vector subspace $g' \subseteq V$ such that $\dim g' = 2$ and $g = \mathcal{P}g'$.

Lemma 9.23. *For any vector space V, the set $\mathbb{P}V$ forms an incidence geometry with $\mathbb{P}$-lines as the lines.*

Lemma 9.24. *In a projective plane over any field, any two different $\mathbb{P}$-lines intersect in exactly one point.*

Definition 9.25. An ***embedding of incidence geometries*** is an injective map $X \hookrightarrow Y$ such that points in X are colinear if and only if their images are colinear. A bijective embedding of incidence geometries is known as a ***colineation***.

Lemma 9.26. *Any injective vector space homormophism $W \hookrightarrow V$ induces an embedding $\mathbb{P}W \hookrightarrow \mathbb{P}V$.*

Definition 9.27

Let E be an affine space. Then we can equip the disjoint union

$$\mathbb{V}E := E \sqcup \mathbb{P}\vec{E}$$

with an incidence structure by designating lines to be:

1. Sets $\mathbb{V}g := g \sqcup \{\vec{g}\}$ for any affine line $g \in E$,
2. and $\mathbb{P}$-lines $l = \mathbb{P}l'$.

We call this structure the ***projective completion of E***. We call points in $\mathbb{P}\vec{E}$ ***points at infinity***.

Exercise 9.28. *If E is a two-dimensional affine space, then its projective completion $\mathbb{V}E$ contains exactly one line containing only infinitely distant points. We call this line the* **line at infinity**.

Exercise 9.29. *Given an affine space E, the trivial embeddings $E \hookrightarrow \mathbb{V}E$ and $\mathbb{P}\vec{E} \hookrightarrow \mathbb{V}E$ are embeddings of incidence geometries. Any injective affine map $F \hookrightarrow E$ of affine spaces is an embedding of incidence geometries and induces an embedding $\mathbb{V}E \hookrightarrow \mathbb{V}F$ of their projective completions.*

Theorem 9.30

Let V be a vector space. Let E be an affine hyperplane in V not containing the origin. Then

$$\mathbb{V}E \cong \mathbb{P}V$$

Corollary 9.31: Projective lines are projective completions

Let F be a field. Then we have $\mathbb{P}^1 F \cong \mathbb{V}F = F \sqcup \{\infty\}$.

Exercise 9.32. 1. *For any field $\mathbb{F}$, we have*

$$\mathbb{P}^n\mathbb{F} \cong \bigsqcup_{i=0}^{n} \mathbb{F}^i$$

2. *For a finite Field $\mathbb{F}_q$, we have*

$$\left|\mathbb{P}^n\mathbb{F}_q\right| = \sum_{i=0}^{n} q^i$$

Chapter 10

Hyperbolic Geometry

10.1 Circle Inversions

Definition 10.1. Let E be a Euclidean plane. Then the corresponding ***extended Euclidean plane*** is the set

$$\hat{E} = E \cup \{\infty\},$$

adding a "point at infinity" to E.

Definition 10.2. We call a subset $L \subseteq \hat{E}$ a ***generalized circle*** if it is either a circle in E or the union of a line in E with $\{\infty\}$. We call the former ***real circles*** and the latter ***extended lines***.

Definition 10.3. Let E be a euclidean plane. Then we assign to each generalized circle $L \subseteq \hat{E}$ the ***circle inversion map***

$$s_L : \hat{E} \to \hat{E},$$

which is given by the ordinary reflection at L with the additional rule $s_L(\infty) = \infty$ if L is an extended line, and by

$$c + \vec{v} \mapsto \begin{cases} c + \left(\frac{r}{\|\vec{v}\|}\right)^2 \vec{v} & \vec{v} \neq \vec{0} \\ \infty & \vec{v} = \vec{0} \end{cases}$$

with $\infty \mapsto c$ if L is a circle with center c and radius r.

Exercise 10.4. *s_L is an involution, i.e. we have $s_L^2 = \mathrm{id}$.*

Exercise 10.5. *For any two points $p, q \in \hat{E}$, there exists a circle inversion s_L with $s_L(p) = q$ and $s_L(q) = p$*

Exercise 10.6. *Every circle inversion maps generalized circles to generalized circles.*

10.2 Möbius Transformations

Definition 10.7. A composition of circle inversions is known as a ***Möbius transformation***.

Exercise 10.8. *The set* $\text{Möb}(\hat{E})$ *of Möbius transformations of a given extended Euclidean plane* $\hat{E}$ *forms a group under composition.*

Exercise 10.9. *Any two generalized circles in a generalized Euclidean plane can be mapped onto each other by a Möbius transformation.*

Theorem 10.10. *Given an extended Euclidean plane* E*, its Möbius transformations are precisely the bijections* $\varphi : \hat{E} \xrightarrow{\sim} \hat{E}$ *mapping generalized circles to generalized circles.*

Theorem 10.11. *Any Möbius transformation that leaves a given generalized circle* L *pointwise unchanged is either the circle inversion* s_L *at that circle or the identity map.*

Exercise 10.12. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$ *and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi \circ s_L \circ \varphi^{-1} = s_{\varphi(L)}$

Exercise 10.13. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$*, and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi(L) = L$ *if and only if* $\varphi \circ s_L = s_L \circ \varphi$*.*

Definition 10.14. Let $\hat{E}$ be a generalized Euclidean plane. Given a pair of generalized circles $L, M \subset \hat{E}$ we call L ***orthogonal to M***, which we write $L \perp M$, if $L \neq M$ and $s_L(M) = M$.

Theorem 10.15. *Let* $L, M \subset \hat{E}$ *be generalized circles in a generalized Euclidean plane* $\hat{E}$ *such that* $L \perp M$*. Let* $\varphi : \hat{E} \to \hat{E}$ *be a Möbius transformation. Then* $\varphi(L) \perp \varphi(M)$*.*

Proposition 10.16. *Let* E *be a Euclidean plane,* $g \subset E$ *a line,* $H \subset E \setminus g$ *an open Half-plane with* g *as its boundary, and*

$$K_H = \text{Möb}_H(\hat{E}) := \left\{\varphi \in \text{Möb}(\hat{E}) \;\middle|\; \varphi(H) = H\right\}$$

the set of Möbius transformations leaving H *invariant. Then:*

1. *Any Möbius transformation that leaves* H *invariant also leaves the extended line* $\hat{g}$ *invariant.*
2. $\text{Möb}_H(\hat{E})$ *operates transitively on* H*, i.e. any point of* H *can be mapped onto any other by a Möbius transformation* $\varphi \in \text{Möb}_H(\hat{E})$*.*
3. $\text{Möb}_H(\hat{E})$ *is generated by the set of circle inversions* s_L *with* $L \perp \hat{g}$*.*
4. *Different elements of* K_H *are still different when restricted to* H*.*

10.3 Complex Möbius Geometry

Definition 10.17. We denote by γ the "double complex conjugation map"

$$\begin{aligned}\gamma : \mathbb{C}^2 &\to \mathbb{C}^2\\ (w, z) &\mapsto (\overline{w}, \overline{z})\end{aligned}$$

Theorem 10.18. *The set of all antilinear automorphisms of $\mathbb{C}^2$ is given by*

$$\mathrm{GL}\left(\mathbb{C}^2\right)\gamma := \left\{F \circ \gamma \,\middle|\, F \in \mathrm{GL}\left(\mathbb{C}^2\right)\right\}$$

Definition 10.19. We denote by

$$\mathrm{GL}\left(\mathbb{C}^2\right)\langle\gamma\rangle := \mathrm{GL}\left(\mathbb{C}^2\right) \sqcup \mathrm{GL}\left(\mathbb{C}^2\right)\gamma$$

the set of all automorphisms of $\mathbb{C}^2$ that are either linear or antilinear.

Since $\mathbb{P}^1\mathbb{C}$ consists of subsets of $\mathbb{C}^2$, every map $\mathbb{C}^2 \to \mathbb{C}^2$ mapping lines to lines acts on $\mathbb{P}^1\mathbb{C}$ in a trivial way. By theorem 9.31 $\mathbb{P}^1\mathbb{C} \cong \hat{\mathbb{C}}$, the map

$$\begin{aligned}\iota_{\mathbb{C}} : \hat{\mathbb{C}} &\xrightarrow{\sim} \mathbb{P}^1\mathbb{C} \hookrightarrow \mathbb{C}^2\\ z &\mapsto (z, 1)\\ \infty &\mapsto (1, 0)\end{aligned}$$

is a bijection, which means we also get an action on $\hat{\mathbb{C}}$.

Theorem 10.20: Action of $\mathrm{GL}\left(\mathbb{C}^2\right)\langle\gamma\rangle$ on $\hat{\mathbb{C}}$

The canonical action gives us a group homomorphism

$$\begin{aligned}\varphi : \mathrm{GL}(2;\mathbb{C})\langle\gamma\rangle &\to \mathrm{Ens}^\times(\hat{\mathbb{C}})\\ \gamma &\mapsto (z \mapsto \overline{z})\\ M := \begin{pmatrix} a & b\\ c & d\end{pmatrix} &\mapsto \left(z \mapsto \begin{cases}\frac{az+b}{cz+d} & z \neq \infty \wedge cz+d \neq 0\\ \frac{a}{c} & z = \infty \wedge c \neq 0\\ \infty & z \neq \infty \wedge cz+d = 0\\ \infty & z = \infty \wedge c = 0\end{cases}\right)\end{aligned}$$

Proof. 1. Let $z \in \mathbb{C}$. Translating directly from $\mathbb{C}$ to gives $\mathbb{C}^2$ $\iota_{\mathbb{C}}(z) = \binom{z}{1}$, which means $M(\iota_{\mathbb{C}}(z)) = \binom{az+b}{cz+d}$.

   (a) If $cz + d$, then this point lies on the same line through the origin as $\binom{\frac{az+b}{cz+d}}{1}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point $\frac{az+b}{cz+d}$.

   (b) If $cz + d = 0$, then this point lies on the same line through the origin as $\binom{1}{0}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point ∞.

2. If $z = \infty$, then $\iota_{\mathbb{C}}(\infty) = \binom{1}{0}$, which gets mapped to $\binom{a}{c}$, from which we can proceed as before.

□

Theorem 10.21. *Our canonical group homomorphism* $\varphi : \mathrm{GL}(\mathbb{C}^2)\,\langle y\rangle \to \mathrm{Ens}^{\times}(\hat{\mathbb{C}})$ *is a surjective group homomorphism*

$$\varphi : \mathrm{GL}\left(\mathbb{C}^2\right)\langle\gamma\rangle \twoheadrightarrow \mathrm{Möb}(\hat{\mathbb{C}})$$

whose kernel is the group consisting of all scalar multiplies of the identity.

Proof. 1. Since $\mathrm{GL}(\mathbb{C}^2)$ is generated by the diagonal matrices together with the matrices $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathrm{GL}(2, \mathbb{C})\,\langle\gamma\rangle$ is generated by these matrices along with γ. It therefore suffices to show that the images of these matrices are Möbius transformations. Now, according to our previous analysis of the maps generated by φ, these images are as follows:

   (a) A diagonal matrix with entries λ and μ gets mapped to $\frac{\lambda}{\mu} \cdot z$, which corresponds to a scaled rotation.

   (b) S gets mapped to $z \mapsto \frac{1}{z}$, which is the circle inversion at the unit circle composed with a reflection along the real axis.

   (c) T gets mapped to the translation $z \mapsto z + 1$.

   (d) γ gets mapped to the conjugation $z \mapsto \overline{z}$, which is just the reflection along the real axis.

   All of these maps are known Möbius transformations.

2. We now know that the image of φ contains inversion at the unit circle, all translations, and all scalings. Therefore, it contains all inversions along proper circles. It also contains reflection along the extended real axes and all rotations at the origin, so it also contains reflections along extended lines. Thus, the image contains the generators of $\mathrm{Möb}(\hat{\mathbb{C}})$, and therefore the entire group.

3. An antilinear automorphism cannot stabilize every one-dimensional subspace. Thus, antilinear automorphisms cannot be mapped the identity, i.e. they cannot be contained in the kernel.

   Meanwhile, the only linear automorphisms stabilizing every one-dimensional subspace are scalings, i.e. multiples of the identity.

$\square$

Definition 10.22: Projective Linear Groups

Given a vector space V, the ***projective linear group*** $\mathrm{PGL}(V)$ is defined as the quotient group of $\mathrm{GL}(V)$ by the group $Z(V)$ of all scalar multiples of the identity. The ***projective special linear group*** $\mathrm{PSL}(V)$ is analogously defined as the quotient group of $\mathrm{SL}(V)$ by the group $SZ(V)$ of all multiples of the identity by a unit scalar.

Theorem 10.23: Möbius group as a projective linear group

$$\text{Möb}^+(\hat{\mathbb{C}}) \cong \text{PGL}\left(\mathbb{C}^2\right)$$

Definition 10.24. We denote by τ reflection along the axis $\langle e_1 \rangle$, i.e.

$$\begin{aligned} \tau : \mathbb{R}^2 &\to \mathbb{R}^2 \\ \begin{pmatrix} a \\ b \end{pmatrix} &\mapsto \begin{pmatrix} a \\ -b \end{pmatrix} \end{aligned}$$

Theorem 10.25: Action of $\text{SL}(\mathbb{C}^2)$ on $\hat{\mathbb{C}}$

Our surjective group homomorphism $\varphi : \text{GL}(\mathbb{C}^2) \langle \gamma \rangle \twoheadrightarrow \text{Möb}\left(\mathbb{C}^2\right)$ induces a surjective group homomorphism

$$\text{SL}(\mathbb{R}^2) \langle \tau\gamma \rangle \twoheadrightarrow \text{Möb}_{\mathbb{H}}(\hat{\mathbb{C}})$$

whose kernel is $\pm \text{Id}_{\mathbb{R}^2}$.

Theorem 10.26: Conjugations of $\mathbb{H}$ as a projective special linear group

$$\text{Möb}^+_{\mathbb{H}}(\hat{\mathbb{C}}) \cong \text{PSL}\left(\mathbb{R}^2\right)$$

Part V

Foundations of Differential Geometry

Part VI

Numerical Linear Algebra

Chapter 1

Matrix Factorization

Part VII

Numerical Analyis

Chapter 1

Numerical Integration

1.1 Extrapolation

A summed quadrature formula is a function $T(h) : \mathbb{R}_{\geq 0} \to \mathbb{R}$ which gives the integral of a function $f \in C^k([a,b])$ in the limit $h \to 0$. If the error of our quadrature formula is of order h^γ for a $\gamma \in \mathbb{N}$, we have

$$T(h) = T(0) + \varphi(h)$$

with $\varphi(0) = 0$ and $|\varphi(h)| \leq ch^\gamma$ for some $c \in \mathbb{R}$. Taylor expansion of φ at 0 gives

$$\varphi(h) = c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

for some $c_1, c_2 \in \mathbb{R}$ and, i.e.

$$T(h) = T(0) + c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

If we evaluate at $h/2$, we get

$$T(h/2) = T(0) + c_1 h^\gamma + c_2 h^{\gamma+1} + o\left(\frac{h^{\gamma+1}}{2^{\gamma+1}}\right)$$

This lets us eliminate the $c_1 h^\gamma$-term, which gives us

$$T^*(h) := \frac{T(h) - 2^\gamma T(h/2)}{1 - 2^\gamma} = T(0) + c_2 \frac{1 - 2^{-1}}{1 - 2^\gamma} \cdot h^{\gamma+1} + o(h^{\gamma+1}),$$

which is a new summed quadrature formula with an error of $h^{\gamma+1}$, instead of h^γ, which can be a significant improvement for $\gamma \ll 1$.

1.2 Calculating Convergence Order

Given a non-polynomial function $f \in C^k([a,b])$ whose exact integral $I(f)$ is known, we can investigate the errors

$$e_h = \left| I(f) - Q^N(f) \right|$$

for various $h > 0$. Since $e_h \approx c_1 h^\gamma$, we can evauate at two different step sizes $h, H > 0$ and get

$$c_1 \approx \frac{e_h}{h^\gamma} \approx \frac{e_H}{H^\gamma}$$

and thus

$$\gamma \approx \frac{\log(e_h/e_H)}{\log(h/H)} = \frac{\log(e_h) - \log e_H}{\log(h) - \log(H)}$$

We can now try out various pairs h, H, and know that we have obtained the correct $\gamma \in h$ if these different pairs all give nearly identical answers.

1.3 Nonlinear Optimization

Let $U \subseteq \mathbb{R}^n$. Then typical problems include:

1. Find $x^* \in U$ such that $f(x^*) = 0$ for some $f : U \to \mathbb{R}^m$.
2. Find $x^* \in U$ such that $g(x^*) = \min_{x \in U} g(x)$ for some $g : U \to \mathbb{R}$.

In general, neither of these problems have closed form solutions. Instead, approximate solutions are found by iterative algorithms, i.e. we find $(x_k)_{k \in \mathbb{N}}$ with $x_k \to x^*$.

Definition 1.1. A numerical algorithm which returns a sequence $(x_k)_{k \in \mathbb{N}}$ approximating a solution x^* given some initial vector $x_0 \in U$ is known as:

1. ***Globally convergent***, if $x_k \to x^*$ for all $x_0 \in U$.
2. ***Locally convergent***, if there exists some $\varepsilon > 0$ such that $x_k \to x^*$ for all $x_0 \in U$ with $d(x_0, x^*) < \varepsilon$.

Definition 1.2. If $x_k \neq x^*$ for all k, we can define the ***rate of convergence*** of x_k by saying that a locally convergent algorithm has ***convergence order*** $\boldsymbol{\alpha \geq 1}$ if there exists a $q \in \mathbb{R}$ such that the errors $\delta_k = d(x^* - x_k)$ fulfill

$$\limsup_{k \to \infty} \frac{\delta_{k+1}}{\delta_k^\alpha} = q.$$

If $\alpha = 1$, we additionally demand $q \leq 1$.

Proposition 1.3. *If $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is a contraction, the basic fixed point iteration $x_{k+1} = \Phi(x_k)$ is globally linearly convergent.*

Proposition 1.4. *If $\Phi \in C^1(\mathbb{R}, \mathbb{R})$ has a fixed point x^*, then the fixed point iteration $x_{k+1} = \Phi(x_k)$ is locally convergent if $|\Phi'(x^*)| < 1$ and divergent if $|\Phi'(x^*)| > 1$.*

1.4 Computing Roots Numerically

Theorem 1.5. Bisection Algorithm: *Let $f \in C^0([a, b], \mathbb{R})$ with $f(a)f(b) \leq 0$. Set $\varepsilon > 0$. Consider the following algorithm:*

1. *Set $(a_0, b_0) = (a, b)$ and $k = 0$.*
2. *Set $c_k = \frac{a_k + b_k}{2}$ and*

$$(a_{k+1}, b_{k+1}) = \begin{cases} (a_k, c_k) & \text{if} f(a_k)f(c_k) \leq 0 \\ (c_k, b_k) & \text{otherwise} \end{cases}$$

3. *Return b_{k+1} if $d(b_{k+1}, a_{k+1}) \leq \varepsilon$. Otherwise set $k \leftarrow k + 1$ and return to step 2.*

Then iteration stops after $J \leq 1 + \log_2\left(\frac{b-1}{\varepsilon}\right)$ steps, the interval $[a_J, b_J]$ contains a root, and if $\varepsilon = 0$, then the sequence $(c_k)_{k\in\mathbb{N}}$ converges to a root linearly with $q = \frac{1}{2}$.

Theorem 1.6. Secant Method: *Let $f \in C^0([a, b] \to \mathbb{R})$. Then the following algorithm converges to a root of f, if one exists:*

1. *Let $\varepsilon > 0$, $x_0 = a$, $x_1 = b$, and $k = 1$.*
2. *If $f(x_k) \neq f(x_{k-1})$, set*
$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} f(x_k)$$
3. *Return x_{k+1} if $d(x_{k+1}, x_k) \leq \varepsilon$. Otherwise set $k \leftarrow k + 1$ and return to step 2.*

If $f \in C^1(U \subseteq \mathbb{R}^n, \mathbb{R}^n)$, Taylor approximation gives

$$0 = f(x^*) = f(x) + Df(x)(x^* - x) + \varphi(d(x^*, x)) \approx f(x) + Df(x)(x^* - x),$$

from which we can set $x := x_k$ to get a new iteration

$$x_{k+1} = x_k - Df(x_k)^{-1} f(x_k) \approx x^*$$

as long as $Df(x_k)$ is regular. This is the (multidimensional analogue of) the Newton method, and is generally much more efficient than the secant method.

Theorem 1.7. *Let $f \in C^2(U, \mathbb{R}^n)$. Let $x^* \in U$ be a root of f such that $Df(x^*)$ is invertible. Then there exists an $\varepsilon > 0$ such that the Newton method converges for any $x_0 \in U$ with $d(x^*, x_0) < \varepsilon$. Additionally, there exists a constant $c \geq 0$ such that the iterates $(x_k)_{k\in\mathbb{N}}$ fulfill*

$$\|x^* - x_{k+1}\| \leq c\|x^* - x_k\|^2$$

If $d(x^*, x_0) \geq \varepsilon$, the Newton method might diverge. In this case, introducing a damping factor $\omega \in (0, 1)$ and iterating

$$x_{k+1} = x_k - \omega Df(x_k)^{-1}$$

can lead to convergence, but makes convergence asymptotically slower (no longer quadratic in general).

Part VIII

Complex Analysis

Part IX

Point-Set Topology

Chapter 1

Topological Spaces

Chapter 2

Filters and Convergence

Theorem 2.1. *Let X be a topological space. Let $U \subseteq X$. Then U is open iff. U is contained in every ultrafilter converging to some point in U.*

Proof. $\Rightarrow$: By definition, if U is an open neighborhood of x, any filter convering to x must contain U.

$\Leftarrow$: Let $U \subseteq X$ and assume $U \in \mathcal{U}$ for every ultrafilter converging to a point $u \in U$. Assume U is not open. Then there exists a point $u \in U$ such that every open neighborhood of u contains at least one point in $X \setminus U$. But then any two members of the family

$$\{X \setminus U\} \cup \{U(u)\}$$

have nonempty intersection. Therefore, this family can be extended to an Ultrafilter $\mathcal{U}_u$. Since $\mathcal{U}_u$ contains all open neighborhoods of u by construction, it converges to u, and must thus contain U. Since it also contains $X \setminus U$, we arrive at a contradiction.

□

Definition 2.2. Let $\mathcal{F}$ be a filter on X. Let $f : X \to Y$. Then the ***pushforward Filter of f on $\mathcal{F}$*** is the set

$$f_*(\mathcal{F}) := \left\{A \subseteq Y \,\middle|\, p_i^{-1}[A] \in \mathcal{F}\right\},$$

Exercise 2.3. *Verify that the pushforward of a filter is a filter, and that the pushforward of an ultrafilter is an ultrafilter.*

Theorem 2.4: Characterization of continuity in terms of Ultrafilters

Let $f : X \to Y$. Then f is continuous if and only if $f_*(\mathcal{U})$ converges to $f(x)$ for every ultrafilter $\mathcal{U}$ that converges to $x \in X$.

Proof. 1. Assume f is continuous. Let $\mathcal{U}$ be an ultrafilter on X converging to $x \in X$.

Let $U \subseteq Y$ be an open neighbourhood of $f(x)$. Then, since f is continuous, $f^{-1}(U)$ is an open neighbourhood of x. We thus have $f^{-1}(U) \in \mathcal{U}$, which gives us $U \in f_*(\mathcal{U})$ by the definition of the pushforward filter.

Thus, every open neighbourhood of $f(x)$ is contained in $f_*(\mathcal{U})$, so $f_*(\mathcal{U})$ converges to $f(x)$.

2. Assume that the pushforward of any ultrafilter converging to x converges to $f(x)$. Let $U \subseteq Y$ be open.

   Let $\mathcal{U}$ be an ultrafilter on X converging to some point $x \in f^{-1}(U)$. Then, by our hypothesis, $f_*(\mathcal{U})$ converges to $f(x)$. Since U is an open neighborhood of $f(x)$, we have $U \in f_*(\mathcal{U})$, and thus $f^{-1}(U) \in \mathcal{U}$. Thus, by theorem 2.1, $f^{-1}(U)$ is open, so f is continuous.

□

Theorem 2.5. *Let $(X_i, \mathcal{O}_i)_{i\in I}$ be a family of topological spaces. Let $\mathcal{F}$ be a filter on $\prod_{i\in I} X_i$. Then $\mathcal{F}$ converges to a point $x \in \prod_{i\in I} X_i$ if and only if each pushforward filter $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$.*

Proof. $\Rightarrow$: Assume that $\mathcal{F}$ converges to $x \in \prod_{i\in I} X_i$. Since all projections p_i are continuous, each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$ by theorem 2.4.

$\Leftarrow$: Assume that each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$. By definition of the product topology, the sets $S_i := p_i^{-1}(U_i)$ for open $U_i \in X_i$ form a subbase of $\prod_{i\in I} X_i$. Therefore, any open neighborhood $U \in \mathcal{U}(x)$ contains an open neighborhood $B = \bigcap_{i\in J\subseteq I} p_i^{-1}(U_i)$ for some open neighborhoods $U_i \in \mathcal{U}(x_i)$.

Since each $(p_i)_*(\mathcal{F})$ converges to x_i, we have $U_i \in (p_i)_*(\mathcal{F})$ for each i. Therefore, by definition of the pushforward filter, we have $p_i^{-1}(U_i) \in \mathcal{F}$ for all i. Since filters are closed under intersection, we get $B \in \mathcal{F}$, and since $B \subseteq U$ we also get $U \in \mathcal{F}$ as desired.

□

Chapter 3

Separation Axioms

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic",
Steven Vickers

Definition 3.1: Separation Axioms

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$:

1. ***T0***, if for every pair of points $x, y \in X$, one of them has a neighbourhood that isn't a neighbourhood of the other.

2. ***T1***, if for every pair of points, both of them have neighbourhoods not containing the other.

3. ***T2*** or ***Hausdorff***, if for every pair of points, both of them have disjoint neighbourhoods not containing the other.

4. ***regular***, if every point can be housed orf from any closed set not containing it by disjoint open sets.

5. ***T3*** or ***regular Hausdorff***, if it is regular and Hausdorff.

6. ***completely regular***, If every point can be separated from any closed set not containing it by a continuous function:

$$\begin{aligned}
&\forall (X \setminus C) \in \mathcal{O} : \\
&\forall x \notin C : \\
&\exists f \in C(X, [0,1]_{\mathbb{R}}) : \\
&\quad f(x) = 1,\ f[A] \subseteq \{0\}
\end{aligned}$$

7. ***T3a***, ***T3$\frac{1}{2}$***, or ***completely regular Hausdorff***, if it is completely regular and Hausdorff.

8. ***normal***, if every pair of closed sets can be housed orf from each other by disjoint neighbourhoods.

9. ***Normal Hausdorff***, if it is normal and Hausdorff.

Chapter 4

Compactness

The familiar notions of open converings and compactness generalize directly from metric spaces to topological spaces. The most major difference is that sequential compactness is generally weaker than "covering compactness", and has to be replaced with a notion of compactness based on ultrafilter convergence.

Definition 4.1. Let $(X, \mathcal{O})$ be a topological space. Then we call a set $\mathcal{U} \subseteq \mathcal{O}$ of open subsets an ***open covering*** of X if every point of X is contained in the union of all elements of $\mathcal{U}$, i.e. if

$$\bigcup_{U_i \in \mathcal{U}} U_i = X$$

Definition 4.2: Compactness

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***compact*** if every open covering of X contains a finite open covering as a subset. We call a subset $K \subseteq X$ compact if it is compact as a topological space equipped with the subspace topology.

Exercise 4.3. *Every closed subset of a compact space is compact.*

Exercise 4.4. *Let $(X, \mathcal{O})$ be compact and $f : X \to Y$ be continuous. Then $f[X]$ is compact.*

Exercise 4.5. *Let $(X, \mathcal{O})$ be a nonempty topological space. Then the following are equivalent:*

1. *$(X, \mathcal{O})$ is compact, i.e. every open covering has a finite subcovering.*
2. *Every set $\mathcal{A}$ of closed sets whose intersection is the empty set has a finite subset whose intersection is the empty set.*
3. *If $\mathcal{A}$ is a set of closed sets such that the intersection over any finite subset of $\mathcal{A}$ is non-empty, the intersection of $\mathcal{A}$ is non-empty.*
4. *Every filter on $(X, \mathcal{O})$ has a limit point.*
5. *Every ultrafilter on $(X, \mathcal{O})$ is convergent.*

Theorem 4.6. **(Alexander Subbase Theorem)** *Let $(X, \mathcal{O})$ be a topological space. Let $\mathcal{S}$ be a subbasis. Then X is compact if every open covering of X by sets of $\mathcal{S}$ has a finite subcovering.*

Proof. Assume that X is not compact. Then there exists a non-convergent ultrafilter $\mathcal{F}$, i.e. we have

$$\begin{aligned} &\forall x \in X: \\ &\exists U_x \in \mathcal{U}(x): \\ &\ (X \setminus U_x) \in \mathcal{F} \end{aligned}$$

By definition of subbases, every $U_x \in \mathcal{U}(x)$ contains an open subset O_x of the form

$$O_x = \bigcap_{k=0}^{n} S_{k,x}$$

for some $S_{k,x} \in \mathcal{S}$ and $n \in \mathbb{N}$. We therefore have

$$\begin{aligned} X \setminus U_x &\subseteq X \setminus O_x \\ &= X \setminus \bigcap_{k=0}^{n} S_{k,x} \end{aligned}$$

which implies $X \setminus \bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ by the filter axioms. Since the intersection of this set with $\bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ is empty, and filters are closed under finite intersection, $\mathcal{F}$ cannot contain every $S_{k,x}$. Therefore, since $\mathcal{F}$ is an ultrafilter, it must contain at least one set $X \setminus S_{k,x}$ for every x. We will refer to these $S_{k,x}$ with $X \setminus S_{k,x} \in X$ as S_x.

Since every S_x is an open neighborhood of x, and thus contains x, the set of all S_x is an open convering of X. By compactness, there must be a finite subcovering

$$\bigcup_{x \in X'} S_x = X.$$

Now, since $\mathcal{F}$ contains every $X \setminus S_x$, it also contains

$$\bigcap_{x \in X'} (X \setminus S_x) = X \setminus \bigcup_{x \in X'} S_x = X \setminus X = \emptyset.$$

which contradicts $\mathcal{F}$ being a filter. □

Lemma 4.7. *Let $(X, \mathcal{O})$ be a Hausdorff space. Let $K \subseteq X$ be compact. Then K can be separated from any point $x \in X \setminus K$ by open neighborhoods, i.e:*

$$\begin{aligned} &\forall x \in X: \\ &\exists U_x \in \mathcal{U}(x), U_K \in \mathcal{U}(K): \\ &\ U_x \cap U_K = \emptyset \end{aligned}$$

Exercise 4.8. *Let $(X, \mathcal{O})$ be compact and Hausdorff. Let $A, B \subseteq X$ be closed. Then A*

and B can be separated from each other by disjoint open neigborhoods.

Exercise 4.9. *Every compact subset of a Hausdorff space is closed.*

Exercise 4.10. *Every continuous function from a compact space into a Hausdorff space is closed.*

Exercise 4.11. *Every continuous injective function from a compact space into a Hausdorff space is an embedding.*

Theorem 4.12: Tychonoff's Theorem

Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Then if all X_i are compact, $\prod_{i \in I} X$ is compact.

Proof.

1. If there exists an $i \in I$ such that $X_i = \emptyset$, then $\prod X_i = \emptyset$ is compact.

2. Now, assume $\prod_{i \in I} X_i \neq \emptyset$. Then there exists an ultrafilter $\mathcal{U}$ on $\prod_{i \in I} X_i$.

   Since each X_i is compact, each pushforward $(p_i)_*(\mathcal{U})$ converges to an $x_i \in X_i$. Thus, by theorem 2.5, $\mathcal{U}$ converges to $(x_i)_{i \in I}$ and $\prod_{i \in I} X_i$ is compact.

□

Chapter 5

Topologies on Algebraic Structures

5.1 Topological Groups

Definition 5.1

Let $(G, \circ)$ be a group. Let $\mathcal{O}$ be a topology on G. Then we call $(G, \circ, \mathcal{O})$ a ***topological group*** iff. the maps $^{-1} : g \to g^{-1}$ and $\circ : G \times G \to G$ are continuous.

Exercise 5.2. *Show that $(G, \circ, \mathcal{O})$ is a topological group if and only if the map*

$$\varphi : (g, h) \mapsto g \circ h^{-1}$$

is continuous.

Exercise 5.3. *Let $(G, \circ, \mathcal{O})$ be a topological groups. Then for any $g \in G$, the left and right group operation maps $l_g : h \mapsto g \circ h$ and $r_g : h \mapsto h \circ g$ are homeomorphisms.*

Exercise 5.4. *Let $(G, \circ, \mathcal{O}_G)$ be a topological group. Let $(H, \circ, \mathcal{O}_H)$ be a normal subgroup equipped with the supspace topology, and let $(G/H, \circ, \mathcal{O}_q)$ be the quotient group, equipped with the quotient topology. Then the quotient map $p : G \to G/H$ is continuous and open.*

Corollary 5.5. *If H is closed in G and G is Hausdorff, then G/H is Hausdorff. If G/H is T1, then H is closed.*

Part X

Algebraic Topology

Chapter 1

Homotopy

Definition 1.1

Let X, Y be topological spaces. Let $f, g : X \to Y$ be continuous. Then f is called ***homotopic*** to g via a map h iff. h is a "continous interpolation between f and g, i.e:

1. h is a continous map $X \times [0, 1] \to Y$,
2. For all $x \in X$, we have $h(x, 0) = f(x)$,
3. For all $x \in X$, we have $h(x, 1) = g(x)$.

We call h a ***free homotopy***, or simply a ***homotopy***, between f and g. We write $f \sim_h g$ if f is a homotopy between f and g. We write $f \sim g$ if there exists a homotopy between f and g.

Exercise 1.2. *Show that the property of being homotopic is an equivalence relation.*

Definition 1.3. We say that f and g are ***homotopic relative to a subset*** $\boldsymbol{A \subseteq X}$ and write $f \sim_{\text{rel } A}$ or $f \sim_{h,\text{rel } A}$ iff:

1. $f \sim_h g$,
2. For all $t \in [0, 1]$, we have $h(t, \cdot)|_A = f|_A = g|_A$

I personally find the designation "homotopic relative to A" quite unintuitive, given that the property of being homotopic relative to a given subset is stronger than the property of being homotopic.

Exercise 1.4. *Show that* $f \sim_h g$ *is equivalent to* $f \sim_{h,\text{rel } \emptyset} g$.

Exercise 1.5. *Let* $f_0, f_1 : X \to Y$ *and* $g_0, g_1 : Y \to Z$ *be continuous. Let* h_0 *be a homotopy between* f_0 *and* f_1 *and let* h_1 *be a homotopy between* f_1 *and* g_1. *Then* $H : (x, t) \mapsto h_1(h_0(x, t), t)$ *is a homotopy between* $g_0 \circ f_0$ *and* $g_1 \circ f_1$.

Definition 1.6

Let $f : X \to Y$ be continuous. Then f is called a ***homotopy equivalence*** between X and Y iff. there exists a map $g : Y \to X$ such that $g \circ f$ is homotopic to id_X and $f \circ g$ is homotopic to id_Y. We write $X \simeq_{f,g} Y$, $X \simeq_f Y$, or simply $X \simeq Y$.

Exercise 1.7. *Show that the property of being homotopy equivalent is an equivalence relation.*

Exercise 1.8. *Show that homeomorphism implies homotopy equivalence.*

Definition 1.9. A topological space X is called ***contractible*** iff. X is homotopy equivalent to a one-point space.

Exercise 1.10.

1. *Show that*

$$h : (x, t) \mapsto x - tx$$

is a homotopy between $\mathrm{id}_{\mathbb{R}^n}$ *and* $f : \mathbb{R}^n \mapsto \{0\}$.

2. *Conclude that* $\mathbb{R}^n$ *is contractible.*

3. *Conclude that homotopy equivalence does not imply homeomorphism.*

Exercise 1.11. *A topological space X is contractible if and only if there exists a point* $x_0 \in X$ *such that* id_X *is homotopic to the constant map* $X \to x_0$.

Definition 1.12. Let X be a subset of a topological vector space. Then we call X ***star-shaped*** iff. there exists a point $x_0 \in X$ such that X contains the connecting line between x_0 and any other point $x \in X$, i.e.

$$\begin{aligned} \forall x \in X &: \\ \forall t \in [0,1] &: \\ (1-t)x_0 + tx &\in X \end{aligned}$$

Exercise 1.13. *Show that any star-shaped set is contractible. Conclude that any convex set is contractible.*

Lemma 1.14. *Let X be a topological space. Let $A \subseteq X$. Then A and X are homotopy equivalent if there exists a homotopy between* id_X *and* $\mathrm{id}_X|_A$, *i.e. a continuous map* $h : X \times [0,1] \to X$ *such that for all $x \in X$:*

1. $h(x, 0) = x$,

2. $h(x, 1) \in A$,

3. $x \in A \implies h(x, 1) = x$

1.1 The Fundamental Group

Definition 1.15. Let X be a topological space. Then a continuous map $\gamma[0,1] \to X$ with $\gamma(0) = \gamma(1)$ is known as a ***loop***. We denote the set of all loops in X with $\gamma(0) = \gamma(1) = x_0$ as $\Omega(X, x_0)$.

Lemma 1.16. *Let $\gamma : [0,1] \to X$ be a loop. Let $\varphi : [0,1] \to [0,1]$ be a continuous map such that $\gamma(0) = 0$ and $\gamma(1) = 1$. Then $\gamma \circ \varphi$ and γ are homotopic relative to $\{0,1\}$.*

Proof. The map

$$h : (x,t) \mapsto \gamma((1-t)x + t\varphi(x))$$

is a homotopy between γ and $\gamma \circ \varphi$. □

Definition 1.17. Let $f, g \in \Omega(X, x_0)$. Then we define the ***concatenation of f and g*** as the map

$$(f * g)(t) = \begin{cases} f(2t) & 0 \leq t \leq 1/2, \\ g(2t-1) & 1/2 \leq t \leq 1. \end{cases}$$

Definition 1.18. Let $\gamma \in \Omega(X, x_0)$. We define the ***inverse loop*** of γ to be the loop that "moves through gamma backwards", i.e.

$$\overline{\gamma}(t) := \gamma(1-t)$$

Exercise 1.19. *Verify that $\overline{\overline{\gamma}} = \gamma$.*

Exercise 1.20. *Let $\gamma \in \Omega(X, x_0)$. Then $\gamma * \overline{\gamma}$ is homotopic to $e_{x_0} : X \to \{x_0\}$ relative to $\{0,1\}$.*

Theorem 1.21: Existence of the fundamental group

Let X be a topological space. Let $x_0 \in X$. Then the quotient space

$$\pi_1(X, x_0) = \Omega(X, x_0)/\sim_{\text{rel }(0,1)}$$

of loops modulo homotopy becomes a group, known as the ***fundamental group of*** (X, x_0), when equipped path concatenation

$$[f] * [g] = [f * g],$$

as the group operation. The neutral element of this group is $e_{x_0} := (x \mapsto x_0)$.

Theorem 1.22. *If we additionally define*

$$\pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$$
$$[\gamma] \mapsto [f \circ \gamma]$$

for all continuous $f : X \to Y$, the map π_1 becomes a covariant functor from the category of topological spaces into the category of groups.

Definition 1.23. Whenever we need to make the base point clear, we write $\pi_1(f, x) := \pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$

Theorem 1.24: Fundamental groups at different base points

Let X be a topological space. Let $x, y \in X$. Then every continuous path $\delta : [0,1] \to X$ induces a group isomorphism

$$\begin{aligned} J_\delta : \pi_1(X, \delta(0)) &\cong \pi_1(X, \delta(1)) \\ [\gamma] &\mapsto \left[\overline{\delta} * \gamma * \delta\right] \end{aligned}$$

Proof.

1. Since loop concatenation preserves homotopy, J_δ sends different representatives of the same equivalence class to the same equivalence class.
2. J_δ goes from $\pi_1(X, x)$ to $\pi_1(X, y)$, since for any $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1) = x$, $\overline{\delta}$ traces out a path from $\delta(1)$ to $\delta(0)$, then γ traces out a loop from $\delta(0)$ to $\delta(0)$, and then δ traces out a path from $\delta(0)$ back to $\delta(1)$, i.e. $\overline{\delta} * \gamma * \delta$ is a loop from $\delta(1)$ to $\delta(1)$.
3. J_δ is a group homomorphism, since

$$\begin{aligned} J_\delta([\gamma_1 * \gamma_2]) &= \left[\overline{\delta} * \gamma_1 * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * e_{x_0} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta * \overline{\delta} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta\right] * \left[\overline{\delta} * \gamma_2 * \delta\right] \\ &= J_\delta([\gamma_1]) * J_\delta([\gamma_2]) \end{aligned}$$

   (Recall that any map that respects the group operation also respects the neutral element and inverses, and is thus a homomorphism.)

4. J_δ is bijective with $J_\delta^{-1} = J_{\overline{\delta}}$:

$$\begin{aligned} J_\delta(J_{\overline{\delta}}(\gamma)) &= \left[\overline{\delta} * \left(\delta * \gamma * \overline{\delta}\right) * \delta\right] = [\gamma] \\ J_{\overline{\delta}}(J_\delta(\gamma)) &= \left[\delta * \left(\overline{\delta} * \gamma * \delta\right) * \overline{\delta}\right] = [\gamma] \end{aligned}$$

□

Corollary 1.25. *If X is path connected, $\pi_1(X, x) \cong \pi_1(X, y)$ for any $x, y \in X$.*

Definition 1.26. Let $\gamma_0 \in \Omega(X, x)$ an $\gamma_1 \in \Omega(X, y)$. Then we call $h : [0,1] \times [0,1] \to X$ a ***free homotopy of loops***, or simply ***loop homotopy***, between γ_0 and γ_1 iff. h is a homotopy between γ_0 and γ_1 such that $h_t(x) := h(x, t)$ is a loop for any fixed $t \in [0,1]$.

Exercise 1.27. 1. *Verify that h is a loop homotopy iff. h is a homotopy and $h_t(0) = h_t(1)$ for any $t \in [0,1]$.*

2. *Show that any $\{0,1\}$-homotopy is a loop homotopy.*

The additional constraint that and $h(\cdot, t)$ is a loop may seem superfluous, but there are in fact homotopies between loops that aren't loop homotopies:

Exercise 1.28. 1. *Let $\gamma_0 = \gamma_1 : [0,1] \to \{0\}$ be constant loops. Let $h(x,t) := xt(1-t)$. Then h is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

2. *Let $\gamma_0(x) = e^{2\pi i x}$. Let $\gamma_1(x) = e^{4\pi i x}$. Then $h(x,t) := e^{2\pi i(1+t)x}$ is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

3. *Assume that any loop $\gamma : [0,1] \to S^1$ with 1 as its base point is homotopic a loop of the form $\gamma(x) := e^{2\pi n i x}$ for some $n \in \mathbb{Z}$ (we will prove this later). Does 2. imply that the fundamental group $\pi_1(S^1, 1)$ is trivial?*

Definition 1.29. A loop $\gamma \in \Omega(X,x)$ is called ***contractible in X*** iff. there exists an $x \in X$ and a loop homotopy h such that $\gamma \sim_h e_x$.

Lemma 1.30. *$\gamma \in \Omega(X,x)$ is contractible in X if $\gamma \sim_{\text{rel } 0,1} e_{\gamma(0)}$.*

Theorem 1.31. *Let $\gamma \in \Omega(X,x)$ be a loop. Let $S^1 \cong I/\{0,1\}$ be equipped with the quotient topology. Then γ is contractible if and only if $\gamma \circ p : S^1 \to X$ has a continuous extension to the closed unit ball $\overline{B}^2$.*

Theorem 1.32. *Let h be a homotopy between $f, g : X \to Y$. Let $x_1 \in X$. Then*

$$\begin{aligned} h(x_1, \cdot) : [0,1] &\to Y \\ t &\mapsto h(x_1, t) \end{aligned}$$

is a path from $f(x_1)$ to $g(x_1)$ which fulfills

$$\pi_1(g, x_1) = J_{h(x_1,\cdot)} \circ \pi_1(f, x_1)$$

Recall that $\pi_1(g, x_1)$ is the map $[\gamma] \mapsto [g \circ \gamma]$, and that J_δ is the map $[\gamma] \mapsto [\overline{\delta} * \gamma * \delta]$, so this theorem states that

$$[g \circ \gamma] = \left[\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)\right]$$

for any $\gamma \in \pi_1(X, x_1)$, or, equivalently, that the diagram

$$\begin{tikzcd} & \pi_1(X, x_1) \arrow[dl, "\pi_1(f, x_1)"'] \arrow[dr, "\pi_1(g, x_1)"] & \\ \pi_1(Y, f(x_1)) \arrow[rr, "J_{h(x_1,\cdot)}"] & & \pi_1(Y, g(x_1)) \end{tikzcd}$$

commutes.

Proof. Define

$$H_t(x) := h(x_1, t + (1-t)x).$$

We have $(f \circ \gamma) \in \Omega(X, f(x_1)$ and $(g \circ \gamma) \in \Omega(X, g(x_1))$, since

$$\begin{aligned} f(\gamma(0)) = f(x_1) = f(\gamma(1)) \\ g(\gamma(0)) = g(x_1) = g(\gamma(1)) \end{aligned}$$

Then a homotopy between $g \circ \gamma$ and $\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)$ is given by:

$$k(x,t) := (\overline{H_t} * (h(\cdot, t) \circ \gamma) * H_t)(x),$$

since we get:

$$\begin{aligned} k(x,0) &= (\overline{H_0} * (h(\cdot,0) \circ \gamma) * H_0)(x) \\ &= (\overline{H_0} * (f \circ \gamma) * H_0)(x) \\ &= (\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot))(x) \end{aligned}$$

and

$$\begin{aligned} k(x,1) &= (\overline{H_1} * (h(\cdot,1) \circ \gamma) * H_1)(x) \\ &= (h(x_1, (1-1) + (1-(1-1))(\cdot)) * (g \circ \gamma) * h(x_1, 1 + (1-1)(\cdot)))(x) \\ &= (h(x_1, \cdot) * (g \circ \gamma) * h(x_1, 1))(x) \\ &= (e_{g(x_1)} * (g \circ \gamma) * e_{g(x_1)})(x) \\ &= (g \circ \gamma)(x), \end{aligned}$$

where the second to last step follows since the loop $(g \circ \gamma)$ ends on the base point $g(x_1)$. □

Lemma 1.33. *Let $\gamma : [0,1] \to X$ be continuous. Let $f : X \to Y$. Then*

$$\pi_1(f, \gamma(1)) \circ J_\gamma = J_{f \circ \gamma} \circ \pi_1(f, \gamma(0)),$$

i.e. the diagram

$$\begin{tikzcd} \pi_1(X, \gamma(0)) \arrow[r, dashed, "\pi_1(f,\gamma(0))"] \arrow[d, "J_\gamma"] & \pi_1(Y, f(\gamma(0))) \arrow[d, "J_{f\circ\gamma}"] \\ \pi_1(X, \gamma(1)) \arrow[r, dashed, "\pi_1(f,\gamma(1))"] & \pi_1(Y, f(\gamma(1))) \end{tikzcd}$$

commutes.

Proof. Let $\sigma \in \Omega(X, \gamma(0))$. Then we have

$$\begin{aligned} \pi_1(f, \gamma(1)) \circ J_\gamma([\sigma]) &= \pi_1(f, \gamma(1))([\overline{\gamma} * \sigma * \gamma]) \\ &= [f \circ (\overline{\gamma} * \sigma * \gamma)] \\ &= [(f \circ \overline{\gamma}) * (f \circ \sigma) * (f \circ \gamma)] \\ &= \left[\overline{(f \circ \gamma)} * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= J_{f \circ \gamma}([f \circ \sigma]) \\ &= J_{f \circ \gamma} \circ \pi_1(f, \gamma(0))([\sigma]) \end{aligned}$$

□

Theorem 1.34. *Let $f : X \to Y$ be a homotopy equivalence. Let $x_0 \in X$. Then the map $\pi_1(f, x_0)$ is a group isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.*

Chapter 2

Fibrations and Coverings

Definition 2.1: Locally Trivial Fibration

Let $\pi : Y \to X$ be continuous. Then π is called a ***locally trivial fibration*** iff. for every $x \in X$, there exists an open neighborhood U such that there exists a homeomorphism $h_u : \pi^{-1}[U] \cong U \times \pi^{-1}[x]$ fulfilling $p_1 \circ h_U = \pi|_{\pi^{-1}[U]}$.

$\pi^{-1}[x]$ is called the ***Fiber of π at x***, and sometimes written F_x.

$$\begin{tikzcd} Y \supseteq \pi^{-1}[U] \arrow[r, "h_U"'] \arrow[d, "\pi"] & U \times \pi^{-1}[x] \arrow[dl, "p_1"] \\ X \supseteq U & \end{tikzcd}$$

Definition 2.2. Let $\pi : Y \to X$ be a locally trivial fibration. Then:

1. If all $\pi^{-1}[x]$ are homeomorphic, we call π a ***fiber bundle***.

2. If every $\pi^{-1}[x]$ is discrete, we call π a ***covering map***.

3. If we can pick $U = X$, π is called a ***trivial fibration***.

Exercise 2.3. *If we replace $\pi^{-1}[x]$ in the definition by an arbitrary topological space F_U, we get $F_U \cong \pi^{-1}[x]$. Thus, both definitions are equivalent.*

Lemma 2.4. *Let $\pi : Y \to X$ be a locally trivial fibration. Let X be connected. Then π is a trivial fibration.*

Proof. Let $x_0 \in X$. Let $V = \{x \in X \mid F_x \cong F_{x_0}\}$ be the set of all points whose fibers are homeomorphic to the fiber of x_0. Let U be a neighborhood as in the definition. Then the fibers of all $x \in U$ must be homeomorphic by definition. Thus, every point in V has a neighborhood contained in V, so V is open. We have

$$X \setminus V = \bigcup_{y \notin V} \left\{x \in X \,\middle|\, F_x \cong F_y\right\} := \bigcup_{y \notin V} V_y$$

Since every V_y is open by the same argument, $X \setminus V$ must be open. But since X is connected with $(X \setminus V) \cup V = X$ and V nonempty, $X \setminus V$ must be empty. □

Theorem 2.5. *Let $\pi : Y \to X$ be continuous. Then the following are equivalent:*

1. *π is a covering map.*
2. *For every $x \in X$, there exists an open neighborhood U of x and a non-empty I such that for ecah $i \in I$, there exists an open set V_i such that:*
   (a) *$\pi^{-1}[U] = \bigcup_{i \in I} V_i$.*
   (b) *For $i \neq j \in I$, we have $V_i \cap V_j = \emptyset$.*
   (c) *For all $i \in I$, the restriction of π to V_i is a homeomorphism between V_i and U.*

Definition 2.6: Lifting

Let $\pi : Y \to X$ be a locally trivial fibration. Let $f : Z \to X$ be continuous. Then a continuous map $\hat{f} : Z \to Y$ is called a ***lift from f to Y*** if $\pi \circ \hat{f} = f$.

Any path can be lifted:

Theorem 2.7. *Let $\pi : Y \to X$ be a locally trivial fibration. Let $\gamma : [0,1] \to XX$ be continuous. Let $\pi(y_0) = \gamma(0)$. Then there exists a lift $\hat{\gamma} : [0,y] \to Y$ with $\hat{\gamma}(0) = y_0$ such that $\pi \circ \hat{f} = f$.*

And lifts respect fundamental groups:

Theorem 2.8. *Let $\pi : Y \to X$ be a locally trivial fibrations. Let $f : Z \to X$. Let $\hat{f} : Z \to Y$ be a lift of f. Then we have:*

$$\begin{aligned}
&\forall z_0 \in Z : \\
&\exists y_0 \in Y : \\
&\quad \hat{f}(z_0) = y_0, \\
&\quad \pi_1(f)[\pi_1(Z, z_0)] \subseteq \pi_1(\pi)[\pi_1(Y, y_0)]
\end{aligned}$$

Proof.

$$\begin{aligned}
\pi_1(f)[\pi_1(Z, z_0)] &= (\pi_1(\pi \circ \hat{f}))[\pi_1(Z, z_0)] \\
&= (\pi_1(\pi) \circ \pi_1(\hat{f}))[\pi_1(Z, z_0)] \\
&=
\end{aligned}$$

□